Three Reductions for Certified Unknot Hierarchies

Cancellation clusters, unsaturated interval bundles,
and polynomial kernels for repeated boundary patterns

Research continuation prepared for Vladimir Reshetnikov
ProveIt project: mathematical results, certificate implementations, and evidence

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Abstract

We develop three complementary reductions for the ProveIt unknot-recognition program. First, normal vertex-disc discovery can be organized by connected cancellation profiles. If at most k edges carry cancellation, these edges form at most r tetrahedron-incidence components, and each edge meets at most d tetrahedra, discovery takes $t^{r+O(1)}(Cd^2)^k \operatorname{poly}(k, B)$ bit operations in a one-vertex rank-one source. The proof includes the exceptional homogeneous case. A certified 15-tetrahedron meridian shows that cancellation support need not be connected.

Second, an orientable interval bundle over a bordered surface B, attached along a nonempty proper subset A of its boundary, adds exactly $a-\chi(B)$ genuine three-dimensional one-handles, where a counts attached intervals. This gives a nonabelian free-product formula, an exact handlebody criterion, and a native classifier extending 30 completed cases to 35 in a retained 106-case corpus. Saturated bases contribute precisely one residual two-handle apiece; their attaching curves remain a separate obligation.

Third, connected-component routing extends the incoming four-copy marked-cut kernel. Every tile whose components meet both interfaces has an aligned power of exponent at most its number d of components, with a sharp bound. A binary power of an explicit size-Q tile consequently has a context-preserving threshold-three query kernel of size at most $6d^2Q$, constructed without expanding its exponent or a global permutation period. Independent replay retains original cut-edge addresses. We also determine the exact universal connected-tile cutoff at every nonnegative threshold.

Complete proofs, executable prototypes, negative controls, paired measurements, and a prioritized research program accompany the report. These are local advances toward quasi-polynomial recognition; the required global geometric, discovery, pattern-transport, and restart bounds are stated explicitly.

Source checkpoint. ProveIt revision `a28a49ee5c41`, including the six relevant incoming research archives at that revision. The supplied code is an additive research package.

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1 Research position and the advances established here

The useful objective is to reduce the work of a complete geometric recognizer while preserving a verifiable path from each computation to the original topological input. Binary normal coordinates already remove the need to list every normal disc. They do not, by themselves, bound the search for a useful surface, the cost of constructing the next patterned manifold, or the number of unsuccessful hierarchy attempts. The three reductions in this report address these three interfaces at different points.

The search result replaces a parameter counting arbitrary exceptional edges by a more informative pair: their total number and their number of connected regions. The geometric result identifies a large class of interval-bundle attachments for which actual topology is determined by a relative one-handle construction. The pattern result eliminates large repetitions even when a tile initially has disconnected, mismatched component partitions. Each result has a concrete producer/checker implementation or an explicitly identified mathematical extension. None is used to infer a global theorem whose hypotheses have not been proved.

1.1 The maintained baseline and the incoming reports

The inspected repository checkpoint, labelled “New research reports,” is

`a28a49ee5c4145db0facbeb1e7f65cd8a2816775`.

It includes a substantial maintained recognizer, many preceding research modules, and the six relevant incoming packages listed below [1, 2, 3, 4, 5, 6, 7, 14]. The present archive does not replace the maintained recognizer or change its default dispatch policy.

Prior component	Available at the checkpoint	Contribution of this report
Cancellation and edge-space discovery	Complete bounded-support ray search; polynomial positive vertex-disc gate	Connected local profiles, component-sensitive complexity, disconnected meridian witness
Geometric reductions	Certified local changes and source provenance	A parameterized search theorem compatible with certified source changes
Marked parallelity	Source interval bundles, ordered interfaces, core proofs; disk-interface census	Proper-boundary annular attachments, free-product classification, five new native completions
Interface homology	Integral condensation with base and attachment data	Geometric strengthening for unsaturated bases; localization of residual relations
Corner-aware boundary kernels	Genuine corner identifications and maximal boundary runs	Attachment counts use actual merged runs and actual owners
Four-copy patterns	Sharp connected repetition kernel; restricted eventual periodicity for disconnected tiles	Independent component routing and bounded reblocking for every through-component tile

In particular, the maintained source-parallelity construction already gives an *inherited* complementary core with at most $30t$ handles for a t-tetrahedron source, independently of the normal multiplicities [2]. This is an important existing geometric result. It is different from a complete construction of an admissible, uniform-type, patterned hierarchy structure with every subsequent curve and disc transported. The latter obligations are preserved here. A raw bounded core must not be silently promoted to that stronger object.

The incoming cancellation bound has the form

$$t^{k+O(1)} \exp(O(k \log(k+1))) \operatorname{poly}(B),$$

where k is a cancellation-support budget. Our bound

$$t^{r+O(1)} (Cd^2)^k \operatorname{poly}(k, B)$$

improves a different part of the parameter range. It can be worse when an edge meets many tetrahedra or the support has many components. There is therefore no reason to discard the earlier method; a parameter-aware dispatch or a cost-accounted race can retain both.

1.2 External mathematical context

The quasi-polynomial objective must be situated in the existing literature. Lackenby's announcement of quasi-polynomial unknot recognition predates this work. His 2026 preprint develops incompressible-surface hierarchies, efficient cutting operations, and certificate constructions; its discussion of hierarchy iterations explicitly distinguishes established procedures from a completed precise running-time analysis [9, 8]. We distinguish the general class $2^{(\log n)^{O(1)}}$ from the stronger target $n^{O(\log n)}$ stated in the February 2021 slides throughout. The present report is an implementation-oriented research continuation, not a claim that quasi-polynomial recognition was first proposed here.

Compressed orbit counting already has the foundational Agol–Hass–Thurston algorithm [10]. The source-parallelity machinery and efficient operations on surface curves should be built on this literature, including Lackenby's 2026 curve algorithms [11]. The difficult interfaces cannot be dismissed by saying that all compressed topological operations are unknown. Conversely, a polynomial algorithm for a specified surface-curve operation does not automatically construct the missing three-dimensional source charts.

The algebra of the component partitions used below is classical: these through-component diagrams are block bijections in the dual symmetric inverse monoid [18, 19]. Terminal cut sparsification is also established theory [15, 16, 17]. Our contribution is the specific marked, binary-power reduction with explicit source lifting, together with its elementary sharp reblocking bound and application to this research interface. The proofs are presented for direct assessment. The search performed for this report does not establish exhaustive priority for their specialized formulations.

1.3 A map of the main statements

The following distinctions are part of the claims, not implementation details. The cancellation theorem promises a disc when a suitable vertex disc lies in the specified class; failure to find one is not a nontrivial-knot certificate. The bundle theorem proves actual topology from actual embedded attachments; an arbitrary incidence spreadsheet is not such an attachment. The graph theorem preserves a terminal query; it does not preserve the entire graph up to homeomorphism or certify the surrounding manifold to be a ball.

Result	Exact mathematical content	Remaining interface
Theorem 3.3	Complete discovery in a specified (k, r, d) class with polynomial bit arithmetic	Prove useful parameter bounds on sources actually reached by recognition
Theorems 4.2, 4.3	Relative one-handles; free factors; exact handlebody genera	Transport marked curves and source charts through the reduction
Corollary 4.5	One residual two-handle per saturated base	Encode and process its genuine attaching curve
Theorem 5.4	$5dQ$ marked kernel for aligned routing	Produce a faithful descriptor from geometric data
Theorem 6.3, Corollary 6.4	Sharp linear reblocking; $6d^2Q$ kernel for all through tiles	Handle components missing an interface and unrestricted nesting

The report gives complete local proofs, a separate finite replay path for each positive output class, and retained reproducible evidence. It does not contain a formal proof-assistant verification. That distinction matters particularly for the native geometric adapters, whose accepted source proofs remain part of the trusted mathematical and software interface.

2 Encoded normal coordinates and the cancellation model

We work with a finite triangulation given by explicit tetrahedron face pairings. Tetrahedra may have identified faces and edges; local edge occurrences are retained before passing to global edge classes. The source validator checks the manifold and boundary hypotheses required by each operator. Numerical surface coordinates and multiplicities use exact binary integers. Bit length, not numerical value, is the size charged for such a quantity. The source tetrahedron count is t, while n later denotes the original knot-diagram input size. These parameters are not interchangeable without a proved conversion and hierarchy-size bound.

2.1 Why cancellation is a nonnegative even increment

In the one-vertex, rank-one setting, choose a primitive integral cocycle c. An essential properly embedded disc in an orientable manifold is coorientable. Its essential boundary on the unique torus is nonseparating, and a curve running along that torus can be pushed to cross the disc once. The dual cohomology class of the disc is consequently primitive. Since the first Betti number is one, that class is $[c]$ or $-[c]$.

An oriented edge has some positive and some negative intersections with the disc. Their signed difference is the value of its dual cocycle, while their sum is its geometric edge weight. In a one-vertex triangulation every vertex coboundary is zero. The representative is therefore exactly c or $-c$ on every edge, and

$$w_e = |c_e| + 2z_e, \qquad z_e \in \mathbb{Z}_{\geq 0}.$$

This is the incoming cancellation model [3]. It does not assert that every such increment is a normal surface, a vertex ray, or a disc. Those are separate constraints. With several global vertices, nonzero coboundaries change the representative and couple the local searches; fixing $|c|$ then loses completeness.

2.2 Local reconstruction and its finite hyperplanes

Write w_{ij} for the six local edge weights of a tetrahedron with vertices $0, 1, 2, 3$. The corner number in a face ijk at i is

$$a_{i;jk} = \frac{w_{ij} + w_{ik} - w_{jk}}{2}.$$

For a normal surface these are nonnegative integers. Of the three opposite-edge sums

$$w_{01} + w_{23}, \qquad w_{02} + w_{13}, \qquad w_{03} + w_{12},$$

the maximum is attained at least twice. Choosing which smaller sum is allowed specifies the quadrilateral sector. The four triangle counts are the minima of the three corner numbers at each tetrahedron vertex. The remaining corner differences specify the single permitted quadrilateral count. These formulas uniquely reconstruct admissible standard coordinates from the local edge weights; on a sector boundary the zero quadrilateral types agree.

For example, if the permitted quadrilateral avoids edges 01 and 23 and has count q, those edges have weights $x_0 + x_1$ and $x_2 + x_3$, while each other edge ij has weight $x_i + x_j + q$. The opposite-edge sums are then $\sum_i x_i$, $\sum_i x_i + 2q$, $\sum_i x_i + 2q$. This also proves the four-point description and the triangle-minimum rule directly in each of the three sectors.

On a glued face the three shared global edge weights determine the same three corner numbers on both sides, so normal matching is automatic. The affine sector boundaries are therefore among the edge-zero equations, the three differences of opposite-edge sums, and the twelve corner-zero equations per tetrahedron. Their integer coefficient ℓ^1 norms are at most four even after local occurrences have been identified. These sparse hyperplanes are the source of the local profile theorem.

2.3 Discovery and verification have different costs

A primitive standard normal vector lies on a standard vertex ray if the matching equations restricted to its positive coordinates have nullity one. The incoming gate exhibits a nonzero maximal minor, verifies exact normal coordinates, computes Euler characteristic, and checks boundary cohomology modulo coboundaries. It does not expand the surface into its normal pieces [3]. We reuse that gate unchanged. The new search chooses vectors to present to it; it does not substitute an approximate numerical rank test or a claim based only on Euler characteristic.

In particular, a huge number of triangles in a disc need not make a supplied witness difficult to verify. Finding that witness can still require examining many candidate supports. The component parameter introduced next controls that combinatorial work under a precise existence hypothesis.

3 Discovery by connected cancellation profiles

The cancellation parameter can be refined by retaining the positions of the exceptional edges. The refinement below is independent of any proposed global bound on that parameter. It does not assume that a quadrilateral sector is supplied, and it accommodates repeated local occurrences of a global edge.

Let $\mathcal{T}$ have t tetrahedra and m global edges, one global vertex, and first Betti number one. Assume that its underlying compact connected orientable manifold has exactly one torus boundary. Write c for a primitive integral cocycle and $b_e = |c_e|$. Every essential normal disc

has weights $w = b + u$, where all u_e are nonnegative even integers. Its cancellation support is $F = \{e : u_e > 0\}$.

Define G to have the global edges as its vertices, with two distinct vertices adjacent when their edges occur in a common tetrahedron. Let

$$d = \max_e \#\{\Delta : e \text{ occurs in } \Delta\}, \qquad q(F) = \#\pi_0(G[F]).$$

An edge occurring several times in one tetrahedron contributes one to d, although all its local occurrences are retained in the equations. Thus the maximum graph degree is at most $5d$. Put $q(\varnothing) = 0$. Both d and q concern this triangulation; neither is invariant under changes of source.

3.1 The local product theorem

Use the global edge-coordinate hyperplanes $\mathcal{H}$: the equations $w_e = 0$, the three differences of opposite-edge sums in each tetrahedron, and the twelve face-corner expressions in each tetrahedron. There are at most $m + 15t$ rows. Every row is supported on at most four global edges, has integer coefficients with sum of absolute values at most four, and comes from one tetrahedron or one edge. Identifications can cancel or combine terms but do not increase this bound.

For a set S of edges let $\mathrm{st}(S)$ be the tetrahedra that meet it, and let $\mathcal{H}_S$ consist of the rows whose restriction to $\mathbb{R}^S$ is nonzero. Then

$$|\mathcal{H}_S| \le |S| + 15|\,\mathrm{st}(S)| \le (1 + 15d)|S|. \tag{1}$$

Theorem 3.1 (Product of local affine profiles)**.** *Let $F_1, \dots, F_r$ be the components of $G[F]$. Fix any admissible quadrilateral sector Q. In the affine space $b + \mathbb{R}^F$, the conditions for membership in its edge-coordinate cone C_Q factor as a Cartesian product of polyhedra on the separate coordinate sets F_i. The ith factor uses only tetrahedra in $\mathrm{st}(F_i)$.*

Suppose $w = b + u$ generates a standard vertex normal ray and that $\operatorname{supp} b \not\subseteq F$. For each i there are $s_i = |F_i|$ hyperplanes from $\mathcal{H}_{F_i}$ such that the system

$$R_i u_{F_i} = -a_i, \qquad (a_i)_\ell = \ell(b), \quad (R_i)_{\ell,e} = \ell_e, \tag{2}$$

is nonsingular and has exactly the target increment as its solution. Its local reconstruction agrees with w on $\mathrm{st}(F_i)$ and with the coherent vector elsewhere.

Proof. No tetrahedron can meet two different F_i, since its edges would join those components in $G[F]$. The edge-coordinate description of normal surfaces uses only local four-point conditions and face-corner inequalities: normal matching is automatic when global edge weights agree. The constraints in a tetrahedron therefore involve at most one group of variable increments. Constraints in an untouched tetrahedron are constant. If that constant violates the chosen sector, the slice is empty; otherwise its nonempty factors have the stated product form. For a sector containing the target the latter alternative applies.

Choose an unchanged edge e with $b_e > 0$. The affine slice fixes $w_e = b_e$, and hence meets the ray $\mathbb{R}_{>0}w$ at just one point. Extremality of that ray implies that w is an extreme point of $C_Q \cap (b + \mathbb{R}^F)$. An extreme point of a nonempty Cartesian product is extreme in every factor: a nontrivial decomposition in one factor would give such a decomposition of the product point.

At an extreme point of a polyhedron in $\mathbb{R}^{s_i}$, the active equalities and inequality boundaries have normals spanning all of $\mathbb{R}^{s_i}$. Otherwise a small displacement in a nonzero common null direction, of either sign, would stay feasible. These active boundaries are all rows from

$\mathcal{H}_{F_i}$. Select s_i independent ones. This gives (2). Restrictions of one global row cannot couple different components, which also supplies a direct block-rank proof. Finally, each local reconstruction is a target tetrahedron or a coherent tetrahedron, so it is admissible and matches globally. No equality of the target and coherent quadrilateral types is used. □

Lemma 3.2 (The exceptional support is connected)**.** *If a nonzero admissible standard normal vector spans an extreme ray, the support of its positive global edge weights is connected in G. Consequently, if $w = b + u$ is a vertex disc with $u_e > 0$ exactly on F and* $\operatorname{supp} b \subseteq F$*, then $G[F]$ is connected.*

Proof. If the positive edge support has several components, restrict w to one component and set every other edge weight to zero. In each tetrahedron the resulting weights are either the original weights or all zero, because a tetrahedron cannot meet two components. Local reconstruction gives either the original standard coordinates or zero. These vectors satisfy matching and the same quadrilateral constraints. They are nonzero and sum to the original vector, giving a nonproportional nonnegative decomposition and contradicting extremality. In the exceptional situation the positive edge support is precisely F. □

The lemma concerns the support of w, not the support of $w - b$. Subtracting a coherent vector can disconnect that support. In particular it supplies no justification for replacing a component budget $q(F) \le r$ by the unrestricted assertion $q(F) = 1$.

3.2 A complete algorithm and its cost

A *profile on S* consists of a positive even integral solution of (2), the standard coordinates reconstructed on $\operatorname{st}(S)$, and its change in Euler characteristic. Every candidate is checked locally before retention. Distinct descriptions that produce the same increments are deduplicated. The parity restriction loses no essential discs: adding even increments to the coherent edge weights preserves even total weight on every face.

Enumerate all connected edge sets S with $1 \le |S| \le k$. For each, enumerate the $|S|$-subsets of $\mathcal{H}_S$, solve the square affine systems, and retain their profiles. Combine at most r profiles whose supports are pairwise nonadjacent in G and whose total size is at most k. Their tetrahedron stars are disjoint, so their local replacement coordinates assemble without conflict. The coherent surface is tested separately.

For an exceptional connected set S containing $\operatorname{supp} b$, additionally choose $|S| - 1$ rows from $\mathcal{H}_S$. Restrict these rows to $\mathbb{R}^S$, append $\sum_{e \in S} w_e = 1$, and solve whenever the square matrix is nonsingular. Keep strictly positive directions and reconstruct a primitive integral normal vector. Recompute its actual increments relative to b before claiming membership in a cancellation class. The usual active-ray argument supplies one of these systems for every exceptional vertex disc. Lemma 3.2 ensures that no disconnected exceptional support is omitted.

Why the positive gate certifies a connected disc. For a primitive admissible integral vector x, let J be its positive standard-coordinate support. If the matching matrix restricted to J has nullity one, each connected component vector is λx for some $\lambda > 0$. Bezout coefficients for $\gcd(x_j) = 1$ show that an integral vector λx has integral λ: if $\sum a_j x_j = 1$, then $\lambda = \sum a_j(\lambda x_j)$. Since the component coefficients sum to one, there is exactly one component. Nonzero boundary cohomology modulo two ensures nonempty boundary. The classification of compact connected surfaces then makes $\chi(x) = 1$ equivalent to a disc. Its boundary is essential because an essential simple torus curve has primitive slope, whereas an inessential one has zero mod-two homology class. The gate computes the class modulo vertex coboundaries, and a nonzero supported

minor supplies independent exact rank evidence. This is the incoming polynomial gate, reused unchanged.

Theorem 3.3 (Cancellation-cluster discovery)**.** *There is a deterministic exact algorithm which, on a source satisfying the stated hypotheses, finds an essential normal disc whenever there is a standard vertex essential disc with*

$$|F| \le k, \qquad q(F) \le r.$$

Writing B for a bound on the binary length of the coherent weights and source indices, its bit complexity is bounded by

$$t^{r+O(1)}(Cd^2)^k \operatorname{poly}(k, B) \tag{3}$$

for an absolute constant C. A polynomial rank-and-boundary vertex-disc gate suffices; no bound in the expanded normal-piece count is needed. The component budget may first be replaced by $\min\{r, k\}$. For $r = 0$ only the coherent case is promised. The bound may always be combined, by an ordinary race or by comparing certified input parameters, with the previous unrestricted cancellation-support bound.

Proof. For a promised nonexceptional disc, Theorem 3.1 places the true increment on every component among the retained profiles. These profiles have pairwise nonadjacent supports, and the combination stage reconstructs the entire disc. The exceptional case was covered above. Primitive normalization followed by the exact supported-rank, Euler, and boundary-cohomology gate verifies every reported positive result. On a torus boundary a primitive vertex normal vector of Euler characteristic one and nonzero boundary class modulo two is an essential disc. This gate is polynomial in the encoded source and coordinates.

For counting, a graph of maximum degree Δ has at most $m(4\max\{1, \Delta\})^{s-1}$ connected sets of size s. One elementary enumeration chooses a root, a rooted ordered spanning-tree shape (at most 4^{s-1} choices), and one neighbor for each tree edge; every connected set has such a description. With $\Delta \le 5d$, this is at most $m(20d)^{s-1}$. Equation (1) and $\binom{N}{s} \le (eN/s)^s$ bound the affine systems for one support by $[e(1 + 15d)]^s$. Thus, for some absolute C_0, the number of size-s profiles and the work descriptions producing them are at most $m(C_0d^2)^s$, up to polynomial factors per description. The exceptional systems have the same bound: for $s \ge 2$ use $\binom{(1+15d)s}{s-1} \le [2e(1 + 15d)]^{s-1}$, and handle $s = 1$ directly.

For $a \le r$ components of total size $s \le k$, there are $\binom{s-1}{a-1}$ ordered positive size compositions. Even before rejecting incompatible supports, their product lists have length at most $m^a(C_0d^2)^s$. Summing over a, s costs a factor at most $\operatorname{poly}(k)2^k m^r$. Each exact solve, assembly and verification costs a fixed polynomial in t, k, B by the bit estimate below. Absorb the absolute exponential constants into C to obtain (3). Enumeration, rejected extensions, local reconstruction, and verification are included in these charges. □

Proposition 3.4 (Anchored systems have short binary solutions)**.** *For an affine profile on s edges, the rational increments have a common denominator of magnitude at most 4^s and numerators of magnitude at most $s4^s \max_e b_e$. In particular their bit length is $B + O(s + \log(s + 1))$. The integral profiles retained by the algorithm and their assembled normal vectors have bit length $B + O(k + \log(k + 1))$ per coordinate.*

Proof. Every row of the square matrix R has Euclidean norm at most four. Hadamard's inequality gives $|\det R| \le 4^s$, and every cofactor has magnitude at most 4^{s-1}. The right-hand side has entries of magnitude at most $4 \max b_e$. The adjugate formula therefore gives the asserted numerator bound. Only the single right-hand side carries the B-bit coherent weights; multiplying B by the system order is unnecessary. Local reconstruction consists of sums,

differences, minima, and division by two. The retained profiles have even integral increments, and disjoint-star assembly takes one profile value per changed edge and one replacement row per changed tetrahedron. Exceptional homogeneous rays have integer cofactor representatives of $O(s)$ bits before reconstruction and primitive normalization, so they obey the same final bound. □

3.3 A certified disc with two separated cancellation components

Start with the incoming eight-tetrahedron coherent obstruction, attach four of its prescribed Fibonacci boundary layers, and perform the 2–3 moves at the currently labelled sites

$$(4,0), \quad (5,0), \quad (12,0).$$

The resulting solid torus has fifteen tetrahedra, seventeen global edges, one vertex, maximum tetrahedron-incidence degree $d = 8$, and Regina isomorphism signature

`pHKzvvPAPQMcbcdlhlhjmmmlnoohqxrorfjidicxak`.

The global-edge identifiers used below are not dense indices from one to seventeen. Its primitive coherent vector has Euler characteristic -7. The source-bound witness supplied in `disconnected_disc.json` is a primitive standard vertex meridian D with

$$w(D) = b + 4\mathbf{e}_{47} + 4\mathbf{e}_{53}.$$

The tetrahedron stars are $\mathrm{st}(47) = \{7, 13\}$ and $\mathrm{st}(53) = \{8\}$, so the cancellation support has two singleton components in G. The witness has 133 normal pieces and Euler characteristic one; the coherent vector has 127 pieces. All coordinates, full face pairings, the move history, and an independently replayable rank-and-boundary certificate are retained.

This is a counterexample to the claim that the cancellation support of every vertex meridian is connected. The new algorithm with $k = 2, r = 1$ exhausts its restricted class without finding a disc, whereas $k = 2, r = 2$ finds D. The incoming complete weight-space algorithm independently recovers D from the supplied support $\{47, 53\}$.

Proposition 3.5 (Exact minimum cancellation support in the retained source)**.** *For this fifteen-tetrahedron triangulation, the minimum cancellation support size among standard vertex meridians is $\kappa_V(\mathcal{T}) = 2$.*

Proof by exact finite algebraic exhaustion. The independently certified vector D supplies the upper bound. The coherent vector has Euler characteristic -7, so support size zero is impossible. Moreover b is nonzero on fifteen of the seventeen global edges. Thus for any singleton $F = \{e\}$ the exceptional case $\operatorname{supp} b \subseteq F$ is excluded. By the active-equation argument in Theorem 3.1, a vertex meridian with this support must occur at an offset

$$\delta = -\frac{\ell(b)}{\ell_e} > 0, \qquad \delta \in 2\mathbb{Z}, \qquad \ell \in \mathcal{H}, \quad \ell_e \neq 0.$$

The script `audit_one_edge_inventory.py` constructs the dense global rows directly from the local corner and opposite-sum formulas, independently of the new profile producer and its sparse hyperplane builder. It obtains 119 distinct rows, 327 nonzero row–edge incidences, and 35 distinct positive even pairs (e, δ). Exact local normal reconstruction rejects 32 pairs. The remaining three are

e	δ	$\chi(b + \delta\mathbf{1}_e)$
42	4	-3
47	4	-3
53	4	-3

and hence none is a disc. This exhausts support sizes zero and one, proving the lower bound. The source hash, all offsets, all three normal vectors, and exact Euler characteristics are retained in `artifacts/one_edge_inventory.json`. □

This finite proof uses the maintained cocycle and coordinate routines and the incoming local normal decoder; it is independent of the new candidate enumeration, not an independent reimplementation of every topological primitive. It is a reproducible exact computation, not a proof-assistant formalization or a separate generic negative-certificate format. It does not rule out one-component cancellation supports of arbitrary size, nor does it require the following full vertex census.

For additional finite evidence, Regina 7.4 standard-coordinate tree traversal and direct double description independently produced the same 821 vertex vectors. Exactly three passed the meridian test; their $(|F|, q(F))$ pairs are $(2,2),(4,2),(4,2)$. Every positive meridian passes the separate exact certificate checker. The completeness of this additional census relies on those external enumeration implementations; the package does not contain an independently checked negative certificate for absence of all possible one-component vertex meridians. The explicit positive counterexample and the restricted algorithmic theorem do not depend on that stronger census conclusion.

3.4 Scope of the improvement

For bounded d, fixed r, and $k = O(\log t)$, (3) is polynomial. For example, bounded d, $r = O(\log t)$, and $k = O((\log t)^2)$ give quasipolynomial discovery. More generally, when B is polynomial in t, the sufficient budget is

$$r \log(t+1) + k \log(d+1) = O((\log(t+1))^2).$$

These implications are conditional on the existence of a suitable vertex disc in a certified source. An edge may meet $\Theta(t)$ tetrahedra, r may equal k, and no theorem here supplies a source family with the required bounds. In the absence of such information, the unrestricted incoming algorithm can be asymptotically better.

The result is restricted to one global vertex and rank-one cocycles. With several vertices, the signed cocycle can vary by vertex coboundaries; independently solving local factors with a fixed vector b is then incomplete. The earlier sign-chamber method or a separately proved treatment of shared cocycle parameters is needed. Boundary cohomology in the positive verifier must still be taken modulo all boundary-vertex coboundaries; a nonzero raw parity vector is insufficient in that more general setting.

4 Unsaturated interval bundles are relative one-handle additions

The incoming marked-parallelity classifier rejects every annular interface, although many such interfaces are topologically harmless. The distinction that matters is whether the *entire boundary of a base surface* is attached. The following result eliminates all nonsaturated bases at once. It is a statement about genuine embedded interval bundles; a table of Euler characteristics does not supply its geometric hypotheses.

4.1 The relative surface-spine lemma

Let B be a connected compact surface with nonempty boundary. Let $A \subseteq \partial B$ be a nonempty compact one-dimensional submanifold, with a interval components and c circle components.

Endpoints are included. Assume that $A \neq \partial B$. Equivalently, if b denotes the number of boundary components, then $c < b$; maximal interval runs have already been merged at their actual source corners. Write

$$m = a + c, \qquad h = a - \chi(B), \qquad \lambda = h - m + 1 = 1 - \chi(B) - c.$$

Lemma 4.1 (Relative surface bands)**.** *The pair (B, A) admits a handle decomposition starting with a collar of A and adding exactly h surface one-handles, with no additional zero- or two-handles. The collar has m components. The one-handles connect these components, so $h \geq m - 1$ and $\lambda \geq 0$.*

Proof. Choose a triangulation in which A is a boundary subcomplex. Since A is proper, choose an edge e in the interior of a complementary boundary interval, subdividing first if necessary. The dual graph of the triangles, using interior triangle adjacencies, is connected. Root a spanning tree at the triangle incident to e.

Collapse that triangle across e. Then process the remaining triangles in an order in which a parent precedes its children in the dual tree, collapsing each triangle across its edge to its parent. Such an edge is free at that point: the parent triangle has been removed and there is at most one other incident triangle. It was not the edge across which the parent collapsed. All chosen edges except e are interior edges of the original triangulation. Thus no edge or vertex of A is removed. After every triangle has been processed the surviving complex is a connected finite graph L containing A.

In a surface, this collapse gives a relative regular-neighborhood presentation: reverse a triangle collapse by enlarging a collar of its surviving two-edge path. This local enlargement is an isotopy of the boundary away from A, so a regular neighborhood of L relative to A is homeomorphic to B with its specified attachment subset.

Take small vertex discs and edge bands for L, while beginning with the collar of A. Any graph vertices outside that collar can be joined to it by a forest containing no cycle after each component of A has been contracted. More explicitly, root one tree at each component of A and extend the trees until every vertex has been reached, assigning each previously unreached vertex to its first reaching tree. Absorb each extra vertex disc along its parent band, in order away from the roots. This cancels every additional zero-handle against one one-handle without joining two components of the initial collar. All remaining bands are surface one-handles; bands may be twisted when B is nonorientable. There are no two-handles.

A collar of an interval has Euler characteristic one and a collar of a circle has Euler characteristic zero. Every band decreases Euler characteristic by one. Consequently its number is $h = \chi(A) - \chi(B) = a - \chi(B)$. A connected result obtained from m initial components requires at least $m - 1$ connecting bands. □

The triangulation used in this proof can be very large. The proof is an existence argument for a terminal topological classification, and does not promise a polynomial-size explicit band system when the genus is binary encoded. The algorithm below computes its invariant count without constructing those bands.

4.2 A geometric relative-handle theorem

Theorem 4.2 (Proper-boundary bundle attachment)**.** *Let $E \to B$ be an interval bundle with orientable total space, and suppose B, A satisfy Lemma 4.1. Let K be a compact three-manifold, and glue $E|_A$ into ∂K by an embedding with the usual compatible corner collars. Suppose the resulting three-manifold $X = K \cup E$ is orientable.*

After absorbing a collar of the interface into K, the pair has a relative handle decomposition consisting of exactly $a - \chi(B)$ three-dimensional one-handles and no handles of other indices. In particular, the assertion holds for arbitrary embeddings of the annular components of $E|_A$ into ∂K; no primitivity, slope, or null-homotopy assumption is needed.

Proof. Over a collar of A, the total bundle is a collar of $E|_A$. This can be seen separately on interval and circle components. The interval case is a product. A boundary circle has an annular collar in B, and the orientation character of B is zero around it. Orientability of E therefore forces trivial fibre reversal there. Thus the total bundle over this collar is $(S^1 \times I) \times I$, with the prescribed attaching annulus as one collar face. Gluing all these collar pieces to K only enlarges its boundary collar, giving a manifold homeomorphic to K. This conclusion does not depend on how the interface annulus sits on that boundary.

Now lift the relative surface decomposition from Lemma 4.1. An individual band is a disc, so its interval bundle is trivial. Its lift is $D^1 \times D^2$, attached along the two discs $\partial D^1 \times D^2$. It is therefore a genuine three-dimensional one-handle. The attaching discs have disjoint collars at each stage, as supplied by the actual base handle decomposition and bundle charts. There are exactly $a - \chi(B)$ such bands.

When a surface band is twisted, the fibre transition can reverse also; the total space remains orientable by hypothesis. These transitions alter attaching parametrizations, not the fact that the lifted piece is a three-dimensional one-handle. The argument is consequently valid for nonorientable bases as well. □

One must distinguish this theorem from a claim that all annular sums of handlebodies are handlebodies. Properness of A is essential. For an unsaturated annulus base, one attached boundary circle gives $h = 0$, so the solid-torus bundle is just an outward collar of the annular interface. For a saturated Möbius base, the vertical annulus represents the square of a solid-torus generator and the theorem does not apply.

4.3 Exact free products and handlebody classification

Suppose that $K = \bigsqcup_j K_j$, and that pairwise disjoint bundles $E_i \to B_i$ are attached along $E_i|_{A_i}$. Assume the complete intersection contract: all intersections with the core are precisely these vertical interfaces, with no unrecorded triple intersections. Attached bases have $A_i \neq \varnothing$ and actual multiplicity one. Assume they are all unsaturated. Let Γ be the bipartite multigraph with a core vertex for every K_j, a base vertex for every attached B_i, and one edge for each connected component of A_i. Parallel edges are retained. Put

$$\lambda_i = 1 - \chi(B_i) - c_i.$$

A connected component C of Γ has cycle rank $\beta_1(C) = |E(C)| - |V(C)| + 1$.

Theorem 4.3 (Nonabelian unsaturated condensation)**.** *For the component X_C indexed by C, there is a free-product decomposition*

$$\pi_1(X_C) \cong \mathop{*}_{K_j \in C} \pi_1(K_j) * F_{r_C}, \qquad r_C = \sum_{B_i \in C} \lambda_i + \beta_1(C).$$

Here F_r denotes the free group of rank r. After choices of basepoint paths, each natural core inclusion corresponds to the inclusion of its free factor. In particular all core fundamental groups inject.

Moreover, X_C is a handlebody if and only if every K_j in C is a handlebody. If their genera are g_j, its genus is

$$G_C = \sum_{K_j \in C} g_j + \sum_{B_i \in C} (1 - \chi(B_i) - c_i) + \beta_1(C). \tag{4}$$

Proof. Apply Theorem 4.2 simultaneously to all attached bases. The result is obtained from its core pieces by $H = \sum_i (a_i - \chi(B_i))$ one-handles. These handles connect precisely the core pieces indexed by C, because each original B_i is connected and its interfaces are nonempty. If q denotes the number of core pieces in C, choose $q - 1$ of the connecting one-handles forming a tree on them. These tree handles give the free product of the core groups. Each of the other $H - q + 1$ handles adjoins a free generator, by van Kampen. But

$$H - q + 1 = \sum_i (1 - \chi(B_i) - c_i) + \Big(\sum_i (a_i + c_i) - q - \#\{B_i \in C\} + 1\Big),$$

and the second parenthesis is $\beta_1(C)$. This proves the group statement with the indicated natural inclusions.

One-handles added to orientable handlebodies yield orientable handlebodies, with genus equal to the sum of the old genera plus $H - q + 1$. This gives (4). Conversely, cutting the resulting handlebody along all the one-handle co-core discs recovers the core pieces with their added collars. Cutting a handlebody along disjoint properly embedded discs produces handlebodies; equivalently such a disc system extends, after disc slides, to a complete meridian system. Hence every recovered core piece is a handlebody. □

This is stronger than an integral-homology decomposition. It explains geometrically why unsaturated bases in the incoming interface-homology formula impose no new relations on the core. No assertion about an explicit transported generating system or a compressed representation of all boundary inclusions follows from the existence of the splitting.

Corollary 4.4 (Exact ball criterion in this class)**.** *Assume the core genera have been certified. Then X_C is a ball exactly when all core pieces in C are balls, C is a tree, and every attached base in C is orientable and planar with exactly one boundary component not attached as a complete circle.*

Proof. All terms in (4) are nonnegative. For an orientable base of genus g and boundary count b, $\lambda = 2g + b - 1 - c$, which vanishes under $c < b$ exactly when $g = 0$ and $b = c + 1$. For a nonorientable base with $k \geq 1$ crosscaps, $\lambda = k + b - 1 - c \geq 1$. The remaining summands vanish precisely as stated. A connected genus-zero handlebody is a ball. □

Unattached bordered bundles are separate handlebodies of genus $1 - \chi(B)$ and can be counted with binary multiplicity. Their components must be included before asserting that the *whole cut* is one ball. Closed bases are outside this operator. A single ball component among several components does not certify the unknot.

4.4 Localizing the remaining two-handles

Corollary 4.5 (One two-handle per saturated base)**.** *Keep the genuine bundle and interface hypotheses, but allow $A = \partial B$. Set $\varepsilon = 1$ in that case and $\varepsilon = 0$ otherwise. The bundle attachment admits a relative handle decomposition with*

$$h_1 = a - \chi(B) + \varepsilon, \qquad h_2 = \varepsilon,$$

and no additional zero- or three-handles. Thus, after all unsaturated attachments have been reduced, at most one additional two-handle is needed for each saturated base.

Proof. The unsaturated case is Theorem 4.2. In the saturated case, remove the interior of a small disc D from B. The punctured surface B° has one additional, unattached boundary circle and $\chi(B^\circ) = \chi(B) - 1$. Its attachment set is still A, with $a = 0$, so its interval bundle contributes $1-\chi(B)$ one-handles. Restoring $D \times I$ attaches along $\partial D \times I$, exactly the attaching annulus of one three-dimensional two-handle. □

For a connected attachment-graph component C, let σ be its number of saturated bases and put

$$r_C = \sum_{B_i \in C} (1 - \chi(B_i) - c_i + \varepsilon_i) + \beta_1(C).$$

Taking all relative one-handles first and then the σ two-handles also gives the exact, generally noncanonical presentation

$$\pi_1(X_C) \cong \left(\underset{K_j \in C}{*} \pi_1(K_j) * F_{r_C} \right) / \langle\langle R_1, \ldots, R_\sigma \rangle\rangle. \tag{5}$$

Indeed, the one-handles give the displayed free product as in Theorem 4.3, and van Kampen adds one normal relation for each two-handle. The original core maps may cease to be injective after imposing these relations. The formula bounds the number of new relators independently of numerical base genus; it does not furnish their words or their embedding data.

This localizes the topological source of new relations, but does not bound the length of the surviving attaching curve. A high-genus saturated base can contribute a long product of commutators or squares. An algorithm using this corollary must retain that curve and its embedding; its binary genus counter is not an attaching-map certificate. The native prototype deliberately leaves these saturated cases unresolved.

4.5 A saturated counterexample with identical first Betti number

Let K be a solid torus with generator x. In its boundary choose an embedded annulus whose core is a primitive torus slope with longitudinal winding number 3. Let E be the orientable interval bundle over a Möbius band. Then E is a solid torus with generator y, while the core of its vertical annulus represents y^2. Glue these two annuli by an orientation-compatible homeomorphism. Van Kampen gives

$$\pi_1(K \cup E) = \langle x, y \mid x^3 = y^2 \rangle.$$

Its abelianization is $\mathbb{Z}$, since $\gcd(3, 2) = 1$. Its group is nonabelian: the assignments $x \mapsto (123)$ and $y \mapsto (12)$ give a surjection onto S_3. The resulting connected manifold is consequently not a genus-one handlebody despite having its first integral homology. The Möbius base is saturated, as required for exclusion from the theorem.

For comparison, use an annulus base and attach only one boundary circle to the same winding-three annulus in K. Its group presentation is $\langle x, y \mid x^3 = y \rangle \cong \mathbb{Z}$, and the relative collar proof identifies the entire resulting manifold with K. The positive conclusion here comes from the collar geometry, not just the group.

4.6 Encoded algorithm, certification, and limits

The finite operator records each core genus, the Euler characteristic and boundary count of each base, its orientation and multiplicity, and the actual component owner of every maximal interval or circle interface. It rejects closed bases and returns `INCONCLUSIVE` whenever an attached base is saturated. It then forms Γ, computes its connected components, and applies

(4); isolated profiles are added as binary counts. A separate checker uses a core-only graph traversal and the independent relative-handle formula

$$G_C = \sum_j g_j + \sum_i (a_i - \chi(B_i)) - q + 1.$$

It verifies the source fingerprint and exact integer types. Producer functions are not invoked during replay.

For s explicitly named pieces, p interfaces, and maximal numerical bit length L, standard graph traversal and binary arithmetic give polynomial bit cost in $s + p + L$. A straightforward implementation can rescan the interface list for each component, giving an adequate $O((s+p)^2)$ structural bound; indexing interfaces by component makes that phase linear. Multiplication of an isolated multiplicity by a genus is charged at its binary cost. Neither numerical genus nor multiplicity is ever an iteration bound.

The native adapter invokes the incoming source-parallelity, ordered boundary-word, and core-handlebody certificate producers. Its verifier replays those source proofs before deriving the finite model. It checks that ports are partitioned exactly once and that all ports in one connected interface have one actual core owner and one actual base owner. Thus a user-supplied incidence table alone does not become a source-normal-cut certificate.

The output is a terminal *unmarked* topological census. The proof provides a homeomorphism and a relative band system by existence, but the implementation does not emit their source charts. Subsequent hierarchy operations needing transported curves, intersection sequences, or a lifted cutting disc require extra representations and bounds. Saturated planar and nonorientable bases still require genuine two-handle or other attachment information. This result removes a concrete transition obstruction; it does not establish a complete quasi-polynomial recognizer.

5 Repetition kernels from aligned component partitions

The four-copy theorem for connected tiles extends to a class of disconnected tiles substantially larger than matchings of individual input and output ports. The correct hypothesis concerns the partitions of the interface indices induced by connected components. In particular, componentwise transport avoids taking the least common multiple of permutation cycle lengths.

5.1 Conventions and the connected primitive

All graphs are finite undirected multigraphs; loops and parallel edge occurrences are retained. Every edge has unit cut cost. A tile has ordered disjoint interfaces

$$L = (\ell_1, \dots, \ell_w), \qquad R = (r_1, \dots, r_w),$$

where every listed vertex is distinct and $w \geq 1$. Serial composition identifies the output of one occurrence with the input of the next, coordinate by coordinate. It identifies vertices, never edges. Private vertices in distinct occurrences remain distinct unless a prescribed serial seam identifies boundary vertices.

For a coloring $c : V(G) \to \{0, 1\}$ write

$$q_G(c) = |\{uv \in E(G) : c(u) \neq c(v)\}|,$$
$$F_G(c) = \{a : \text{an edge has both endpoints colored } a\}.$$

For a threshold $k \geq 0$, the marked signature $\mathsf{S}_k(G)$ is the minimum cut cost for each prescribed coloring of its exposed vertices and each *exact* mark set $F \subseteq \{0, 1\}$, with every value exceeding k and infeasibility represented by the common value $k + 1$.

The incoming four-copy article proves the following primitive [14]. It is recalled to make the dependency of the new result explicit.

Lemma 5.1 (Connected repetition primitive)**.** *For a connected tile W with equal nonempty input and output widths, put*

$$a_k = \begin{cases} 4, & k = 3, \\ k + 2, & k \neq 3. \end{cases}$$

Then, for every positive integer q,

$$\mathsf{S}_k(W^{\circ q}) = \mathsf{S}_k(W^{\circ \min(q, a_k)}).$$

Moreover, a coloring of $W^{\circ a_k}$ with cut cost at most k contains a zero-cost occurrence of W. Such an occurrence is monochromatic and may be duplicated arbitrarily without changing boundary colors, cut cost, or exact edge marks.

The duplication assertion follows because $a_k > k$, every positive-cost occurrence contributes at least one, and connectedness makes a zero-cost coloring monochromatic. The distinction between equality of *minimum* signatures and preservation of every individual coloring remains relevant: compression in the connected four-copy proof can lower an individual coloring's cost; lifting from the kernel by duplication preserves its cost exactly.

5.2 Aligned partitions

Definition 5.2 (Aligned component partitions)**.** Let $C_1, \ldots, C_d$ be all connected components of a tile T. Define

$$A_j = \{i : \ell_i \in C_j\}, \qquad B_j = \{i : r_i \in C_j\}.$$

The tile has aligned component partitions if every A_j and B_j is nonempty and the two unordered collections of subsets of $[w]$ coincide:

$$\{A_1, \ldots, A_d\} = \{B_1, \ldots, B_d\}.$$

There is then a unique permutation σ of $[d]$ satisfying $B_j = A_{\sigma(j)}$ for every j.

The sets in either collection are a partition of $[w]$. The condition is testable directly from the component graph and the port lists. It includes a connected tile ($d = 1$), and the permutation-zero class where every component has exactly one input and one output ($d = w$). It also permits variable component widths. For example, one component can transport a one-element block to a two-element block, provided another component transports the latter block back. Thus $|A_j| = |B_j|$ is *not* required for individual components.

Within component C_j, order its input ports by their original indices in A_j and its output ports by their original indices in $A_{\sigma(j)}$. Its individual input and output widths may differ. Adjacent components in a σ-orbit have matching intermediate interfaces because they use exactly the same index set. A complete orbit word returns to its initial index set and therefore is an ordinary equal-width connected tile.

Lemma 5.3 (Decomposition into component chains)**.** *Suppose T has aligned component partitions. For every $N \geq 1$, its open power is the disjoint union, over $j \in [d]$, of the connected component chains*

$$P_{j,N} = C_j \circ C_{\sigma(j)} \circ \cdots \circ C_{\sigma^{N-1}(j)}.$$

The input ports of this chain are $L|_{A_j}$ and its output ports are $R|_{A_{\sigma^N(j)}}$, with their original coordinate labels.

Proof. At a serial seam, all outputs of C_j have indices in $B_j = A_{\sigma(j)}$, so they are identified with inputs of exactly one component in the next layer, namely $C_{\sigma(j)}$. Every one of those input ports is used. Conversely, each component in the next layer has exactly one predecessor component because σ is a permutation. Consequently distinct chains do not merge or split. Every component occurrence is on exactly one chain, the one whose initial index is $\sigma^{-s}(j)$ if it is C_j in layer s. Each chain is connected because its components are connected and consecutive components are glued along a nonempty interface. The assertion about external ports follows by tracking these index sets through the N layers. □

5.3 The componentwise kernel theorem

Let $\mathcal{O}$ run over the cycles of σ, and put

$$h_{\mathcal{O}} = |\mathcal{O}|, \qquad Q_{\mathcal{O}} = \sum_{j\in\mathcal{O}} (|V(C_j)| + |E(C_j)|), \qquad Q = \sum_{\mathcal{O}} Q_{\mathcal{O}}.$$

For a fixed $j \in \mathcal{O}$, define the full-cycle word and prefix word by

$$W_j = C_j \circ \cdots \circ C_{\sigma^{h_{\mathcal{O}}-1}(j)}, \qquad V_{j,r} = C_j \circ \cdots \circ C_{\sigma^{r-1}(j)}.$$

The word $V_{j,0}$ is empty: it is an identity interface and contributes no new edge or private vertex. This convention is used in formulas; an implementation need not create a zero-exponent block.

Theorem 5.4 (Kernel without a global permutation period)**.** *Let T have aligned component partitions and let $N \geq 1$ be given in binary. For each cycle $\mathcal{O}$, write*

$$N = q_{\mathcal{O}} h_{\mathcal{O}} + r_{\mathcal{O}}, \qquad 0 \leq r_{\mathcal{O}} < h_{\mathcal{O}}.$$

Replace the chain starting at $j \in \mathcal{O}$ by

$$K_{j,N} = W_j^{\circ \min(q_{\mathcal{O}}, a_k)} \circ V_{j,r_{\mathcal{O}}},$$

omitting an empty factor. Give this chain the original exposed input labels A_j and output labels $A_{\sigma^N(j)}$. Let $K(T,N)$ be the disjoint union of these d chains. Then

$$\mathsf{S}_k(T^{\circ N}) = \mathsf{S}_k(K(T,N)).$$

Its explicit size, counting vertices and edge occurrences before any external gluing, is at most

$$(a_k+1)\sum_{\mathcal{O}} h_{\mathcal{O}} Q_{\mathcal{O}} \leq (a_k+1) dQ.$$

In particular the bound is $5\sum_{\mathcal{O}} h_{\mathcal{O}} Q_{\mathcal{O}} \leq 5dQ$ at threshold three, and is bounded by $(k+3)\sum_{\mathcal{O}} h_{\mathcal{O}} Q_{\mathcal{O}}$ for every k. For fixed k the kernel is constructible in polynomial bit time in the explicit input size and $\log(N+1)$. No global order $\mathrm{ord}(\sigma)$ is constructed as a word length or used for expansion.

Proof. By Lemma 5.3,

$$P_{j,N} = W_j^{\circ q_{\mathcal{O}}} \circ V_{j,r_{\mathcal{O}}}.$$

The full-cycle word W_j is connected and has input and output index set A_j, so Lemma 5.1 applies to its repeated factor when $q_{\mathcal{O}} > 0$. If $q_{\mathcal{O}} = 0$, then $N < h_{\mathcal{O}}$ and the entire chain is

the explicitly retained prefix of N components. There is therefore no use of the connected primitive at exponent zero.

Composition with the unchanged prefix preserves the marked signature: compatible boundary colors agree on the seam, cut costs add because edges are disjoint, and exact marks combine by union. The same argument for disjoint union combines the d chain replacements. Saturation at $k+1$ is legitimate because all cut costs are nonnegative. This proves signature equality with all original exposed colors prescribed.

There is a potentially important boundary-label issue. The retained number of component occurrences is $\min(q_{\mathcal{O}}, a_k)h_{\mathcal{O}} + r_{\mathcal{O}}$, which is congruent to N modulo $h_{\mathcal{O}}$. Thus the terminal component index is the original one, $\sigma^N(j)$. More importantly, its actual port labels are retained coordinatewise. Different cycles may have different retained lengths; the output of the construction is not asserted to be a common power $T^{\circ M}$ for one integer M.

Each full-cycle word contains one occurrence of each component in $\mathcal{O}$, so its size is at most $Q_{\mathcal{O}}$. The prefix has size at most $Q_{\mathcal{O}}$. Each of the $h_{\mathcal{O}}$ chains associated with the cycle therefore contributes at most $(a_k+1)Q_{\mathcal{O}}$. Summing gives the stated bound; vertex identifications only reduce it.

Connected components, index partitions, and permutation cycles are computed from the explicit tile. Division of a binary N by an integer at most d, followed by comparison of its quotient with the fixed number a_k, is polynomial in the encoded input size. The construction writes only the bounded number of component occurrences just counted. If graph identifiers are encoded in binary, the bit size adds the usual logarithmic factor for identifiers; it remains polynomial. This also proves the complexity claim. □

Corollary 5.5 (Singleton routes)**.** *If every component has one input and one output, write $\pi(i)$ for the output index of the component containing ℓ_i. Then $T^{\circ N}$ has w connected two-terminal chains, the chain starting at ℓ_i ending at $r_{\pi^N(i)}$. Theorem 5.4 gives a threshold-three kernel of size at most $5wQ$ using the individual cycle lengths of π.*

5.4 Contextual replacement and mixed flat descriptions

Theorem 5.6 (Contextual equivalence)**.** *Suppose an arbitrary finite graph context contains edge-disjoint tile-power macros. Each macro has either a connected tile or aligned component partitions. The context may identify any of the exposed macro vertices with each other or with context vertices, including ports belonging to the same macro. It has no access to private macro vertices.*

Replacing every macro by its kernel preserves the marked signature at threshold k on any exposed context vertices. In particular it preserves the existence of a cut of size at most k whose two shores each contain an edge. Connected macros contribute $O(Q_i)$ at fixed k; an aligned macro contributes $O(\sum_{\mathcal{O}} h_{\mathcal{O}} Q_{\mathcal{O}}) = O(d_i Q_i)$. Thus any finite flat mixture of these macros has a polynomial-size explicit query kernel, independently of the numeric multiplicities.

Proof. Fix a coloring of the context vertices, including every equivalence class of identified exposed vertices. Its restriction prescribes compatible colors on the exposed vertices of every macro. For each choice of exact local mark sets, the optimal global cost is the sum of the context cost and the optimal macro costs; the global mark set is the union of their exact mark sets. Theorem 5.4 and the connected primitive preserve each summand at threshold k. Taking minima over local mark sets and all allowed context colorings proves the assertion.

The argument permits arbitrary identifications among exposed vertices: they merely restrict the allowed colorings. It does not permit an additional edge or gluing at a private vertex,

because the signature did not record the color of that vertex. All sizes add over the explicitly listed macros and context, and quotienting can only reduce vertex count. □

This theorem concerns a *flat* input: a capped tile is explicit. Unrestricted nesting of macro definitions is not covered by the size bound without an additional argument controlling intermediate expansion. The bound also does not by itself validate a spherical embedding, preserve the graph's entire topology, or certify that an ambient 3-manifold is a ball. Its topological use is the threshold-three terminal query on an actual spherical cubic pattern in a separately certified ball.

5.5 Exact source-witness lifting

Theorem 5.7 (Cycle-word run certificates)**.** *A threshold-three marked cut of $K(T, N)$ lifts to a coloring of the original $T^{\circ N}$ with the same cost, exposed colors, and exact marks. For each component chain, its lifted description uses at most four retained full-cycle coloring blocks and one prefix coloring block. One of the full-cycle blocks may carry a binary repetition count. The description and its verification are polynomial in the encoded source and kernel sizes. At most three original cut-edge addresses and two original monochromatic edge addresses suffice to list its cut and its two edge witnesses.*

Proof. If $q_{\mathcal{O}} \leq 4$, the chain was not shortened, and its explicit kernel coloring is already a source coloring. If $q_{\mathcal{O}} > 4$, its kernel contains four copies of the connected word W_j. Their total cost is at most the global cost, at most three. One of these four copies therefore has zero cost and is monochromatic. Replace it by $q_{\mathcal{O}} - 3$ copies with that same monochromatic coloring. Its input and output colors agree, so the seams remain valid. Its cost remains zero, and its monochromatic edge mark is unchanged. All other full-cycle blocks and the prefix are left unchanged.

Do this independently on each component chain. The chains have no internal shared vertices, and their exposed colors are unchanged, so every external identification remains valid. Exact global marks and cost are preserved. This produces the asserted run representation.

An edge in component $C_{\sigma^u(j)}$ of full-cycle occurrence v has original tile-copy index $vh_{\mathcal{O}} + u$, using zero-based indices. An edge in the prefix has original tile-copy index $q_{\mathcal{O}}h_{\mathcal{O}} + u$. Keep its original local edge-occurrence identifier. These formulas locate edges in the source without enumerating omitted copies. A positive-cost run can contain at most three cut edges in total, while a repeated monochromatic run contributes none. Consequently all cut addresses can be listed directly, and a first monochromatic edge for each mark gives the two mark witnesses. The copy indices have $O(\log(N + 1))$ bits. Checking local colors, seams, edge costs, mark unions, and the binary run counts requires only polynomial work. □

For a general fixed threshold k, use a_k full-cycle blocks and the same proof, since $a_k > k$. The two edge-mark witnesses remain sufficient; the cut list has at most k addresses.

5.6 Hypotheses that must survive implementation

Original output transport. Consider the width-two matching with edges $\ell_1 r_2$ and $\ell_2 r_1$. At exponent five, input colors $(0, 1)$ and output colors $(1, 0)$ admit a zero-cost coloring with both marks. Blindly replacing the exponent by four changes the output transport to the identity; those boundary colors then require positive cost. Cyclewise reduction preserves the original labeled output, whereas a single global cap need not.

A partition mismatch really can merge chains. Let a width-three tile have two components, one with edges $\ell_1 r_1, \ell_2 r_1$, the other with edges $\ell_3 r_2, \ell_3 r_3$. Its input partition is $\{\{1,2\},\{3\}\}$ while its output partition is $\{\{1\},\{2,3\}\}$. The first tile occurrence has two components, but the square of the tile is connected: the second occurrence's component containing ℓ_1, ℓ_2 joins the two previous components, and the other second-layer component attaches to the same union. Thus there is no fixed two-chain decomposition. This example disproves that decomposition outside the alignment condition; it does not prove that every possible kernel fails. Section 6 supplies a verified reblocking strategy that applies to this example and, more generally, to every tile whose components all meet both interfaces.

Naive independent strands lose constraints. A connected four-spoke star with boundary leaves ℓ_1, ℓ_2, r_1, r_2 cannot be replaced by two independent paths merely by pairing ℓ_i with r_i. Prescribing $c(\ell_1) = c(r_1) = 0$ and $c(\ell_2) = c(r_2) = 1$ gives zero cost and both marks on the two independent paths, but is not a zero-cost coloring of the connected star. Shared component vertices cannot be duplicated as if they were independent channels.

Reversal and orientation metadata. Reversing a tile means interchanging its input and output lists and reversing every component word. It replaces σ by σ^{-1} and preserves the alignment condition. The same theorem applies to that explicitly reversed tile. Edge directions or fibre-orientation metadata do not affect an undirected cut cost, but any such data needed for geometric provenance must still be transported separately. A negative fibre sign is not an instruction to complement a shore color. If a different formalism imposes color-complementing seams or accesses hidden corners, it is outside the vertex-identification model used here.

Sharpness and bounds not claimed. The cutoff four cannot be improved uniformly in this class, because the class contains every connected tile. For a one-edge tile, a path with both endpoint colors zero needs at least four edges to have both marks: the coloring $0, 0, 1, 1, 0$ realizes cost two on four edges, while three edges cannot realize both monochromatic edge marks with those endpoints. This establishes sharpness of the *connected primitive cutoff*, not optimality of the factor dQ in the routed size bound.

5.7 Relation to existing graph-sparsification results

The preservation of terminal cut information under graph replacement is classical. Hagerup, Katajainen, Nishimura, and Ragde developed mimicking networks for external flow behavior [15]. For thresholded terminal connectivity, Chalermsook and coauthors construct connectivity-c mimicking networks of $O(pc^4)$ edges for p terminals, with algorithms polynomial in the explicit graph size for fixed c [16]. Liu subsequently obtained $\widetilde{O}(pc^3)$ edges with polynomial construction time in the explicit graph size and c [17].

These results are essential context: this work does not claim that small-cut preservation or polynomial-size terminal cut representatives are new. The statement established here is a particular *marked*, binary-multiplicity graph-grammar reduction, constructed without expanding the input power, together with source-coloring lifting. Ordinary terminal minimum-cut values do not automatically determine the exact edge-mark states used by the terminal pattern query. The literature search located no identical cyclewise formulation, but this is not a priority claim.

6 Polynomial reblocking for every tile with through components

The alignment hypothesis can be obtained after a bounded amount of explicit reblocking. The result below applies whenever every connected component of the tile meets both interfaces. It does not require the input and output partitions to agree initially, and it does not require equal input and output widths within an individual component.

At the level of boundary connectivity, these tiles are the classical block bijections, or elements of the dual symmetric inverse monoid [18, 19]. Their composition by stacking diagrams is exactly the connectivity operation used here. The following elementary stabilization proof is included in full; the contribution for this application is its combination with marked-cut repetition kernels and source-address provenance. No priority claim is made for the underlying semigroup stabilization principle.

6.1 Stabilization of the component partitions

Call a tile *through* if every connected component meets both its input and its output interface. For such a tile T, let d be its number of components, and write A_j, B_j for their input and output index sets as before. All A_j and B_j are nonempty, but the unordered partitions need not agree. Every power $T^{\circ n}$ is again through: each component occurrence can be continued through a nonempty interface in both directions until the exposed interfaces of the whole power are reached.

Define a directed seam relation on the original component indices by

$$j \; D \; l \quad \Longleftrightarrow \quad B_j \cap A_l \neq \varnothing.$$

Every vertex has at least one incoming and one outgoing edge in this relation. For an equivalence relation R on $[d]$, define $f(R)$ to be the equivalence relation generated by all pairs (j, j') for which there are l, l' satisfying

$$j \; D \; l, \qquad j' \; D \; l', \qquad l \; R \; l'.$$

The operator f is monotone for inclusion of equivalence relations.

Lemma 6.1 (Finite component stabilization)**.** *For $n \geq 1$, let E_n relate j and j' when their component occurrences in the first layer belong to the same connected component of $T^{\circ n}$. Then*

$$E_1 = \Delta_{[d]}, \qquad E_{n+1} = f(E_n), \qquad E_n \subseteq E_{n+1}.$$

If $E_n = E_{n+1}$ once, all subsequent relations agree. In particular, $E_d = E_{d+s}$ for every $s \geq 0$.

Proof. The relation for one layer is the identity because the C_j are precisely the connected components of T. Decompose $T^{\circ(n+1)}$ as its first layer followed by a power of length n. The latter power connects its first-layer components according to E_n. Gluing along the seam therefore connects two components j, j' of the new first layer exactly by chains of the pairs defining $f(E_n)$. This proves the recurrence.

The identity relation is contained in every equivalence relation, so $E_1 \subseteq E_2$. Monotonicity of f now proves the remaining inclusions inductively. Equality at consecutive terms is a fixed point of the recurrence and hence persists. Every strict inclusion merges at least two equivalence classes. There are initially d classes, so there are at most $d - 1$ strict inclusions before stabilization. Thus E_d is already stable. □

The number of classes of E_n equals the number $\rho(T^{\circ n})$ of components of $T^{\circ n}$, because every component meets the first layer. This is where the through-component hypothesis is essential.

Lemma 6.2 (The rank criterion for alignment)**.** *If S is a through tile, then*

$$\rho(S^{\circ 2}) = \rho(S) \quad \Longleftrightarrow \quad S \text{ has aligned component partitions.}$$

Proof. Let $r = \rho(S)$. Form the bipartite intersection graph with r vertices on either side, one for each component of the first or second copy of S. Join a first-copy component to a second-copy component if their seam index sets intersect. There are no isolated vertices, and its connected components are exactly those of $S^{\circ 2}$.

Such a bipartite graph has at most r components. It has exactly r if and only if each component contains one vertex on each side, so its adjacency relation is a perfect matching. Because both families of seam index sets are partitions, that matching condition says exactly that each output-index set equals the uniquely matched input-index set. This is alignment. □

Theorem 6.3 (Linear reblocking bound)**.** *Every through tile T with d components has an aligned power $T^{\circ m}$ for some $1 \le m \le d$; in particular $T^{\circ d}$ is aligned. If m is the least such exponent and $r = \rho(T^{\circ m})$, then*

$$m \le d - r + 1.$$

For every $n < m$, $\rho(T^{\circ(n+1)}) < \rho(T^{\circ n})$. The bound $m \le d$ is attained for every $d \ge 2$.

Proof. Lemma 6.1 gives $\rho(T^{\circ d}) = \rho(T^{\circ 2d})$. Apply Lemma 6.2 to $S = T^{\circ d}$ to obtain alignment. If $\rho(T^{\circ(n+1)}) = \rho(T^{\circ n})$, the inclusion $E_n \subseteq E_{n+1}$ must be equality. Stabilization then implies $\rho(T^{\circ 2n}) = \rho(T^{\circ n})$, and the rank criterion shows that $T^{\circ n}$ is aligned. Before the first aligned power, the rank must therefore drop at every step. Starting from d and ending at r gives $m - 1 \le d - r$, as required.

For sharpness, use $w = d + 1$ interface indices $0, 1, \ldots, d$ and the sets

$$A_1 = \{0, 1\}, \qquad A_j = \{j\} \ (2 \le j \le d),$$

$$B_1 = \{0, 2\}, \qquad B_j = \{j + 1\} \ (2 \le j < d), \qquad B_d = \{1\}.$$

Choose any connected graph for each component on its prescribed input and output ports, with distinct components disjoint. The seam relation is the directed cycle $1 \to 2 \to \cdots \to d \to 1$ together with the loop $1 \to 1$. For $1 \le n \le d$, the relation E_n has one nonsingleton class

$$\{1, d, d - 1, \ldots, d - n + 2\},$$

interpreted as $\{1\}$ when $n = 1$, and every remaining class is a singleton. This follows immediately by induction from the recurrence for f. Consequently $\rho(T^{\circ n}) = d - n + 1$ through exponent d, where the power becomes connected. An aligned power would have stable rank at all later powers by the component-chain decomposition. Thus no exponent below d is aligned, proving sharpness. □

6.2 A kernel for all through-component powers

Corollary 6.4 (Polynomial marked-cut kernel after reblocking)**.** *Let T be a through tile of explicit size $Q = |V(T)| + |E(T)|$ with d components. For every fixed $k \ge 0$ and binary exponent $N \ge 1$, the open power $T^{\circ N}$ has a context-preserving marked-signature kernel constructible in polynomial bit time in the encoded input size and $\log(N + 1)$.*

Choose the least aligned exponent $m \le d$, put $S = T^{\circ m}$ and $r = \rho(S)$, and write $N = qm + t$, $0 \le t < m$. If $q = 0$, retain the whole power of length $N < m$. Otherwise use

$$K(S, q) \circ T^{\circ t},$$

where the empty suffix is omitted. The explicit size is at most

$$(a_k + 1)r\, mQ + (m-1)Q \le (a_k + 2)d^2 Q.$$

At threshold three this gives the concrete bound $6d^2Q$.

Proof. At most d explicit copies of T suffice to find the first aligned power by serial construction and component analysis. Its size is at most mQ. The identity $T^{\circ N} = S^{\circ q} \circ T^{\circ t}$ is an exact rebracketing before any reduction. Apply the aligned kernel theorem to the first factor and retain the suffix exactly. Its contextual form establishes the marked-signature assertion. Counting retained records gives the displayed bound, and binary division by $m \le d$ plus the polynomial-size construction gives the bit-time claim. The case $q = 0$ never uses a zero-exponent tile. □

The sharper orbit-sensitive bound is $(a_k + 1)\sum_{\mathcal{O}} h_{\mathcal{O}} Q_{\mathcal{O}} + (m-1)Q$, where the cycles and component sizes are those of the explicit tile S. Using the least aligned exponent also gives $rm \le m(d - m + 1)$. These refinements can substantially reduce the conservative quadratic bound.

Original-source witnesses. While explicitly forming $S = T^{\circ m}$, retain for each edge its pair (u, e) consisting of its zero-based original tile-copy index $0 \le u < m$ and original edge-occurrence identifier e. A lifted cut or mark address $(v, (u, e))$ in copy v of S then has original copy index

$$mv + u.$$

For a retained suffix edge at suffix index u, the original copy index is $mq + u$. These formulas compose with the cycle-word run witnesses and preserve the exact cut cost and marks. A checker can reconstruct S from the original tile, replay the finite routing certificate for S, and replay the source coloring using this affine address map. All copy indices require only $O(\log(N+1))$ bits. None of these steps expands the binary exponent.

Cubic terminal use. Reblocking preserves the actual original seams, ports, and edges. If the original closed flat chain satisfies the cubic local-degree checks, then so does its exact rebracketing; the subsequent routing kernel preserves those degrees as before. The geometric limitation remains unchanged: this is a graph-query theorem and requires a separately certified ball for its terminal topological interpretation.

Scope and audit. The reblocking argument does not cover a component missing one of the interfaces. Such components are not elements of the block-bijection model, and the rank recurrence above no longer counts every component of the power. Its extension to those components would require a further proof. The stabilization claim was exhaustively checked on all 7,089 block bijection diagrams of widths one through five (respectively $1, 3, 25, 339, 6721$ diagrams), using explicit graph composition. In every tested unaligned power the next rank strictly dropped. This audit supports implementation; the lemmas above supply the proof for arbitrary width.

7 Exact universal cutoffs for connected tiles

The incoming article established the universal connected-tile cutoff $k + 2$ and improved it sharply to four at threshold three. Its further questions included the optimal cutoffs at other

thresholds. For the same unit-edge model with two exact monochromatic-edge marks, that question has the following complete answer.

Theorem 7.1 (Exact connected-tile stabilization threshold)**.** *For a nonnegative integer k, put*

$$b_k = \begin{cases} 1, & k = 0, \\ 3, & k = 1, \\ 4, & k = 2, \\ k+1, & k \geq 3. \end{cases}$$

For every connected tile T and every positive integer N,

$$\mathsf{S}_k(T^{\circ N}) = \mathsf{S}_k(T^{\circ \min(N, b_k)}).$$

No smaller cutoff works uniformly for all connected tiles in this model.

Upper bound. A connected tile has at least one edge because its nonempty input and output interfaces are disjoint. A zero-cost coloring of any power is monochromatic, has one exact edge mark, and prescribes that same color on both interfaces. This proves the assertion at $k = 0$. For $k = 1, 2$, start with $N > k + 2$ copies and a coloring of cost at most k. Retain all at most k positive-cost copies and at most one zero-cost copy of each color that appeared. The latter copies are monochromatic, so their deletion does not change seam colors, and retaining one of each occurring color preserves the exact marks. At most $k + 2$ copies remain, including at least one zero-cost copy; duplicate that copy to reach length $k + 2$. Conversely, any coloring of $k + 2$ copies of cost at most k has a zero-cost copy that can be duplicated to obtain every larger length. This proves the two remaining small-threshold upper bounds.

Suppose $k \geq 3$ and $N > k + 1$. Fix a coloring of $T^{\circ N}$ with total cost at most k. At most k tile occurrences have positive cost. Every other occurrence has zero cost, is monochromatic, and has identical constant input and output colors. Consequently it can be deleted from its serial position without creating a coloring disagreement at the new seam. Keep all positive-cost occurrences and at most one zero-cost occurrence of each color that appeared. This preserves boundary colors, cost, and exact marks. At least one zero-cost occurrence remains because $N > k$.

Let p be the number of retained positive-cost occurrences and $z \leq 2$ the number of retained zero-cost occurrences. If $p + z \leq k + 1$, duplicate one retained zero-cost occurrence until the length is exactly $k + 1$. The resulting coloring preserves all the data and its cost.

The only exceptional case is $p = k$ and $z = 2$. Each positive-cost occurrence then has cost exactly one. If some positive-cost occurrence supplies a monochromatic-edge mark $a \in \{0, 1\}$, delete the retained zero-cost occurrence of color a. That mark is still supplied by the positive-cost occurrence, and the other retained zero-cost occurrence supplies the other mark. The new length is exactly $k + 1$ and no required data change.

Otherwise every positive-cost occurrence has no monochromatic edge. All edges of the tile are therefore bichromatic in that occurrence. Its cost is one, so the tile has exactly one edge. Connectedness and the disjoint nonempty interfaces force T to be the single-edge tile with one input and one output. The two retained zero-cost occurrences imply that the required exact mark set is $\{0, 1\}$. For either prescribed pair of endpoint colors, there is a path of length at most four with both marks and cut cost at most two: use colors $0, 0, 1, 1, 0$ for equal zero endpoints, its complement for equal one endpoints, $0, 0, 1, 1$ for distinct endpoints, or its reversal. These paths contain a monochromatic edge and may be padded to length $k + 1$ by duplicating it. The cost is at most two, hence at most k, and the boundary colors and exact mark set are the required ones.

We have proved that every feasible source coloring of cost at most k gives a coloring at length $k+1$ of no greater cost with the same boundary colors and exact marks. Conversely, a coloring of $k+1$ copies with cost at most k contains a zero-cost occurrence. Duplicating it produces a coloring at any larger length with exactly the same cost, boundary colors, and marks. These two directions establish equality of all minimum costs up to the threshold and equality of all saturated entries.

The argument preserves minimum signatures; it does not claim that every individual source coloring can be shortened while preserving its exact cost. The single-edge exceptional case is the only step where its cost may decrease. □

Sharpness. At threshold zero, the smallest possible positive cutoff is one. For threshold one, take the single-edge tile with distinct endpoint colors and require both marks. A path of three edges with colors $0, 0, 1, 1$ has cost one and both marks. No path of at most two edges has these properties: it needs one edge of each monochromatic color and at least one crossing edge. Thus $b_1 \geq 3$.

For threshold two, prescribe both endpoints zero and again require both marks. Four edges with colors $0, 0, 1, 1, 0$ suffice. Any such coloring needs an edge of each monochromatic color and at least two crossing edges, so at least four edges are necessary. Hence $b_2 \geq 4$.

Now let $k \geq 3$ and put $B = k + 1$. Form a connected tile T_k of width $2B$ with input vertices $\ell_1, \ldots, \ell_{2B}$, output vertices $r_1, \ldots, r_{2B}$, and edges

$$\ell_i r_i \quad (1 \leq i \leq 2B), \qquad \ell_i \ell_{i+1} \quad (1 \leq i < 2B).$$

Thus the inputs form a path and every output is attached to its corresponding input by one edge. Prescribe color one on the first B input and output coordinates, and color zero on all the remaining coordinates.

In $T_k^{\circ k}$, propagate these colors unchanged along each vertical strand. Exactly the input-path edge joining coordinates B and $B+1$ is cut in each copy. The total cost is k, and the matching edges supply both exact marks.

In contrast, suppose that $T_k^{\circ(k+1)}$ had any coloring of cost at most k with those boundary colors. Some connected tile occurrence would have zero cost and hence be uniformly colored a. For each of the B oppositely colored original input coordinates, follow its matching-edge strand from the exposed input to the corresponding input of that occurrence. If the occurrence is the first one, its monochromatic input condition already contradicts the prescribed colors. Otherwise these are B pairwise edge-disjoint paths, each with oppositely colored endpoints and hence each containing a cut edge. The total cut cost is at least $B = k + 1$, a contradiction.

Therefore the chosen marked-signature entry is at most k at exponent k and is saturated at exponent $k+1$. If a universal cutoff were at most k, both exponents would reduce to the same capped exponent and would have equal signatures. This contradiction proves $b_k \geq k + 1$. □

Consequences and implementation scope. Every routed or reblocked kernel theorem can use b_k in place of a_k. In particular the routed explicit-size coefficient $a_k + 1$ becomes $k + 2$ for all $k \geq 3$. The threshold-three value remains four, so the production cubic-pattern checker and its source witness format require no change. The supplied general-threshold producer uses the exact value b_k in `connected_cutoff`; both aligned and reblocked construction use it. The certificate solver remains the threshold-three cubic-pattern query. The comb examples establish sharpness in the full connected unit-edge graph-tile class, without a claim of optimality for more restrictive cubic or geometrically realizable subclasses.

8 Terminal interpretation and independently replayed certificates

8.1 What the graph query means topologically

The graph query used by the prototypes asks whether a cubic graph has a cut of size at most three with an edge on each shore. Loops count twice toward degree and never cross a cut; parallel edges remain distinct occurrences. For a cubic graph this is exactly a cut with a cycle on each shore. Indeed, if a shore induces a forest with v vertices and a components, then

$$|\delta X| = 3v - 2(v - a) = v + 2a.$$

A forest containing an edge has $v \geq 2$, $a \geq 1$, and hence boundary at least four. The argument explains why the two monochromatic-edge marks carry the needed information at threshold three. At threshold four they do not have the same interpretation.

For an actual cubic pattern $P \subset \partial B^3 = S^2$, essentiality means that every transverse simple closed curve crossing at most three pattern edges bounds, on at least one side, a disc whose pattern is empty, an arc, or a tripod. The incoming primal criterion states that essentiality is equivalent to absence of the marked cut just described [14, 8].

For completeness, a violating curve has a cycle on either side: a forest in a disc with at most three boundary leaves can only be empty, an arc, or a tripod, since a tree with v trivalent interior vertices has $v + 2$ boundary leaves. Coloring graph vertices by side of the curve gives a marked cut of size at most three. Conversely, in a connected plane graph choose a smallest cut having cycles on both shores. Both shores are connected; otherwise the boundary of a cycle-containing component is smaller and the opposite cycle persists. Such a cut is a bond. The boundary of a regular neighborhood of one shore, with complementary regions away from the other shore filled, gives a simple curve crossing exactly its edges. If the cubic graph is disconnected, a regular-neighborhood boundary separates two cycle-containing components at cost zero. This proves the criterion in all represented cases.

Ordinary edge connectivity would give the wrong predicate. The three edges incident to a vertex of K_4 cut off an allowed tripod. A triangular prism also has edge connectivity three, but its three rungs separate two cycles. The marked query distinguishes them. A geometric importer must additionally handle vertex-free circle components and the empty pattern; the closed-chain API here encodes the nonempty cubic part.

8.2 A finite graph certificate engine

On a finite kernel of size K, enumerate pairs of edge occurrences with disjoint endpoint sets. Require one edge to be monochromatic zero and the other monochromatic one. Give the arcs fixing their endpoints capacity four, and replace every non-loop undirected edge by two opposed unit-capacity arcs. An integral flow computation need only find four units of flow or a smaller minimum cut. In the latter case the induced coloring has the required marks. If every anchor pair supports four units, no marked cut of size at most three exists. This is the unchanged incoming finite certificate engine [14].

A negative graph proof lists a shore; replay recounts its exact cut and its two edge marks. A positive graph proof supplies residual augmenting paths reaching four for every disjoint anchor pair. The checker verifies those paths and requires the complete set of pairs. It does not perform max-flow search. With sparse flow construction, the reference producer has a conservative $O(K^3)$ structural bound and $O(K^3 \log(K + 1))$ certificate size. The independent

dense reference replay has an $O(K^4)$ bound. These are adequate polynomial bounds, not claims to optimal algorithms for cyclic connectivity.

In a source chain, private vertices must have degree three. At every serial seam the corresponding output and input degrees must sum to three. The same checks apply within every repeated block and at the final closure. Interfaces are injective and disjoint before the stated seam quotient. These local conditions prove that the actual expanded graph is cubic, including possible loop edges after closure. The reblocking step is an exact rebracketing of that quotient; the routing step retains the degrees at its genuine labeled seams.

8.3 Three replay layers for the pattern kernel

The pattern producer and checker share immutable input types and elementary validation. They use separate component reconstruction and gluing procedures. The producer follows component chains; replay independently identifies components with a disjoint-set structure and assembles fresh-copy quotient vertices. It verifies the finite graph proof on the reconstructed kernel. For a negative result it also verifies local component color masks, consecutive seams, repeated monochromatic cycle words, the exact binary count of original copies, and the complete list of cut edges and monochromatic anchors.

The through-component wrapper adds a preceding exact reblocking layer. Its certificate binds the original chain, a bounded aligned power of each tile, the unchanged remainder, the adapted-source identity, and the nested routing proof. Its checker constructs the bounded power independently and checks the actual component partitions. The affine address maps in Section 6 then recover original block, copy, and edge-occurrence identifiers. Thus a cut on the reduced graph is accompanied by a succinct coloring of the original graph, even when its exponent has thousands of bits.

It is useful to keep the three correctness assertions separate:

1. The original tile grammar and the adapted grammar describe the same finite graph before reduction, with an exact map of edges.
2. The adapted grammar and the bounded kernel have the same marked query signature; source colors for a low cut lift by zero-cost repetition.
3. The finite graph proof establishes the query answer on that kernel.

For a topological conclusion there is a fourth, geometric assertion: the source grammar is the actual spherical pattern on an independently certified ball. No graph certificate supplies it.

8.4 Return values and the native integration contract

The three new operators have intentionally different meanings.

Operator	Successful output	Meaning of other outcomes
Cancellation search	`DISC_FOUND`, with source-bound normal coordinates and vertex-disc proof	Exhaustion only covers the stated class; limits and unsupported sources remain inconclusive
Bundle classification	`COMPLETE`, with exact component/genus census and source proofs	Saturated bases or unresolved cores remain inconclusive
Pattern query	`LOW_CYCLIC_CUT` or `NO_LOW_CYCLIC_CUT`, with proof	Malformed or unsupported large tiles are rejected; neither result alone is a knot verdict

Discovery limits are charged globally within one query, including attempted systems and profile combinations. External cancellation propagates. Replaying a supplied positive certificate does not call the candidate search. Geometry replay checks the underlying source-parallelity, boundary-word, and raw-core proofs before computing the finite census. These choices make every positive claim reviewable without treating an implementation timeout as evidence of a negative topological conclusion.

The proposed repository integration is additive and opt-in. First import the specific incoming dependencies, then add the new cluster and bundle modules, then expose the graph operator at a boundary where its grammar contract is proved. A native recognizer benchmark would be meaningful only after that producer exists. The present measurements report the actual local operations performed; they do not substitute synthetic graph times for knot-recognition times.

9 Completed experiments and what they establish

9.1 Evidence design

The package retains exact sources, positive certificates, source reconstruction records, independent finite oracles, raw timing rounds, and explicit negative controls. Its tests answer different questions: local algebraic completeness on specified finite sources, preservation of a complete marked signature, correctness of graph assembly, correctness of the original-source witness, and native geometric coverage. Their counts should not be added and described as distinct knot instances.

The final quick runner executes 27 routing/reblocking test methods, 14 bundle test methods, and 11 cancellation test methods, then replays the retained graph and native positive certificates. These are focused suites for the additive prototypes. The maintained recognizer's full regression suite and end-to-end native knot benchmarks are outside the reported experiment.

9.2 Implementation and finite evidence for profile discovery

The general component-budget search is implemented in `normal_weight_clusters.py`. It enumerates connected supports, constructs exact local affine profiles, retains admissible positive even increments, and reuses those profiles in disjoint component products. It also implements the dependent-baseline homogeneous exception. Projected rows are deduplicated, and a same-sign test rejects equations that cannot have strictly positive increments. All source conditions are checked. The component budget is capped at $\min(r, k)$, including when the supplied integer r is very large. Global work, support, linear-system, and product limits remain effective through final certificate replay; a limit reports incomplete search.

The supplied vertex gate is an unchanged copy of the incoming rank-and-boundary checker. It verifies a primitive supported normal ray, Euler characteristic one, and essential boundary through exact linear algebra and boundary cohomology. Positive results are replayed by the separate checker. Exhaustion concerns only the stated source and parameter class and yields no unrestricted knot verdict. The overlay leaves the maintained recognizer and its defaults unchanged.

9.2.1 Complete radius-two candidate generation

The paired experiment compares the new profile search with the incoming enumerator over all supports of size at most two. Each timed call includes source and cocycle preparation, complete candidate generation, local normal reconstruction, and actual-cancellation filtering; neither arm includes a topological vertex-disc gate. Candidate sets are checked for exact equality outside the timed blocks. Each source receives one excluded warmup per arm and five measured rounds, with arm order shuffled using a recorded seed. Table 1 reports seconds and ratios of the two arm medians; these are not medians of per-round ratios. Full samples, observed ranges, counters, and hashes are in `artifacts/benchmark.json`.

Source	t	All supports (s)	Profiles (s)	Ratio
Layered obstruction	8	1.189400	0.026278	45.26
Layered obstruction	16	7.213136	0.043151	167.16
Layered obstruction	24	25.963222	0.060250	430.93
Layered obstruction	40	122.938084	0.096843	1269.46
Disconnected-support source	15	4.006094	0.041128	97.40

Table 1: Complete radius-two candidate generation, five measured rounds. The ratio is the all-support median divided by the profile median. Exact resulting candidate sets agree in every case.

The counts make the change in workload concrete. At $t = 40$, the incoming arm examines 904 weight spaces and 156 927 small systems; the profile arm examines 173 connected supports and 4 051 systems. Only two local profiles survive, producing three assembled candidates including the coherent candidate. For the disconnected-support source, the corresponding counts are 154 spaces and 14 942 systems versus 78 connected supports and 1 420 systems. Three reusable local profiles produce five assembled candidates. Maximum tetrahedron-edge incidence is $d = 6$ in the smallest layered source and $d = 8$ in the other sources.

These gains combine component reuse, smaller connected-support enumeration, projected-row deduplication, and the stricter local positive-even-increment filters. The old baseline first searches homogeneous weight spaces and filters the actual cancellation support after reconstruction. Thus the table does not isolate connectedness as the sole cause of the speedup. The source family is deliberately structured, the parameter radius is small, and these finite timings establish neither a corpus-wide speedup nor an asymptotic growth law.

9.2.2 A retained whole-query regression

The existing incoming one-edge specialization is a stronger practical baseline for positive radius-one queries. A second paired experiment therefore compares complete successful queries, including internal positive verification and replay. Both methods find the same certified disc on these sources. Here there is one excluded warmup per arm and seven measured rounds, again with shuffled arm order. The incoming method uses its native component observer, whereas the new method uses the polynomial rank gate. Their end-to-end costs consequently include different verification mechanisms, an intentional part of this integration comparison.

t	Incoming one-edge (s)	Profiles with rank gate (s)	Slowdown
8	0.002818	0.005381	1.91
16	0.004930	0.014919	3.03
24	0.007299	0.034642	4.75
40	0.012197	0.128793	10.56

Table 2: Complete positive radius-one queries, seven measured rounds. The slowdown is the new median divided by the incoming median, not the median of paired sample ratios.

The larger-radius search improvement therefore does not justify replacing the existing one-edge specialization. A practical portfolio should retain it, while adding component profiles for larger budgets. The polynomial bit-complexity guarantee of the rank gate also does not imply lower constants on these small positive instances. All measured regressions are retained in the artifact.

9.2.3 Validation and evidential limits

All eleven focused tests pass. They compare exhaustive candidate sets on actual sources; recover the disconnected witness through the independent incoming weight-space enumerator; exercise the homogeneous exception; compare connected-set enumeration with brute subsets; check global resource limits, external cancellation, and input immutability; replay certificates with discovery routines disabled; reject forged certificates; and exercise exact arithmetic with an 8193-bit right hand side. That last test validates the local arithmetic path, not a complete topological search at that coordinate size.

The retained fifteen-tetrahedron source includes its full face pairings, construction history, independently checked Pachner moves, and positive vertex-disc certificates. Its independently generated dense one-edge inventory proves the finite statement $\kappa_V = 2$ as described in Proposition 3.5. The separate full census contains 821 standard vertex vectors agreed upon by Regina's direct tree and direct double-description algorithms; the three meridians all have two cancellation components. Each positive is independently certified. Completeness of this additional census still relies on those external enumeration algorithms. No generic independently replayable certificate for unrestricted negative assertions is supplied.

The new module, tests, measurement driver, finite-audit scripts, and theorem fragment form an additive research artifact. They establish a proved search guarantee for an explicit source class and measured gains for complete radius-two candidate generation, together with a concrete integration regression. A global quasipolynomial recognizer still requires the structural and source-coverage results stated elsewhere in this article.

9.3 Native coverage and measured costs

The complete retained source corpus contains 106 normal-cut inputs. The previous disk-only classifier certified 30; the new unsaturated classifier certified 35 and preserved every previous census. All 35 complete outputs passed independent source-proof replay. Each of the five newly covered inputs was also cut explicitly by Regina 7.4; its component-by-component `recogniseHandlebody` results agreed exactly. The remaining 71 cases consist of 55 saturated-base cases, 6 raw-cancellation stalls, and 10 cores with three-handles. These are inconclusive outcomes, not non-handlebody verdicts.

Case	New+replay (ms)	Explicit Regina (ms)	Old (ms)
LST(5,7)	51.439	3.025	20.729
Pachner(3,5,1)	51.711	1.215	29.976
Pachner(5,8,1)	82.614	2.037	42.635
Pachner(5,8,2)	208.660	2.774	96.788
LST(3,8), mixed	310.987	1.881	160.733
Layered 1, one copy	2.681	0.670	2.559
Layered 4, eight copies	423.763	24.471	433.749
Layered 8, one copy	1886.583	195.046	1779.949

The first five old results were `INCONCLUSIVE`; their times are not a speed baseline for a completed answer. All new times include producer and source verifier, while Regina times include construction of the explicit cut, component separation, and recognition. Five paired rounds were run after a warmup, with shuffled arm order and garbage collection outside the timers. The three preserved cases had old/new paired median ratios 0.942, 0.997, and 0.908. On these small inputs the native certificate path is slower than the compiled Regina oracle. The observed advance is additional certified coverage; no practical speedup is claimed for these fixtures.

For the abstract proper-boundary model with an orientable genus $g = 2^{16384} + 17$ base, two boundary circles and one attached annulus, the output is one handlebody of genus $2g$. It represents the $2g$ relative one-handles by an exact counter. The encoded input has 4,321 bytes and the formatted certificate has 25,130 bytes under the benchmark serialization. Production and replay together took a median 0.180 milliseconds over five rounds. No explicit band system or large surface was constructed. The finite prototype tests include 700 random assembly comparisons, strict type and source-mutation checks, producer-disabled replay, saturated refusal, nonorientable-base formulas, and multiplicities exceeding 2^{20000}. Fourteen named test methods passed with the minimal pinned native runtime. Timing values are machine-specific; the raw rounds, environment, source corpus hash, and five replayable new source certificates accompany the code.

9.4 Complete signatures, exact source maps, and sharp examples

The main routing audit samples 240 aligned source tiles of widths one through three, retaining loops and multiple edges. Twenty-seven sources have components with unequal input and output widths. An independent frontier coloring oracle evaluates the assembled kernel; the source is evaluated by exact marked-signature multiplication. The two exponents per source give 480 complete-signature comparisons covering 53,760 table entries, all equal. The largest kernel in this audit has 154 edges. Another 60 explicit cubic source verdicts agree with their kernel verdicts.

A separate reviewer audit reconstructs 300 flat chains with shuffled vertex identifiers, shuffled edge records, reversed tiles, multiple macros, and exponents extending beyond 10^{20}. Its producer/checker graph identities agree. On 120 finite shuffled and reversed cubic chains, every available negative witness was expanded to the original source: 105 complete colorings, covering 96,252 actual edge occurrences, reproduce their exact crossing and anchor addresses. Additional targeted chains combine unstable periods two and three, including zero quotients and nonempty remainders in both blocks.

The reblocking audit enumerates all through-component partition diagrams of widths one through five: respectively $1, 3, 25, 339, 6721$, or 7,089 in total. Independent explicit stacking checks alignment by exponent d and verifies 10,656 strict component-count drops before first alignment. The cubic sharp family in the wrapper tests attains first alignment $m = d$ for

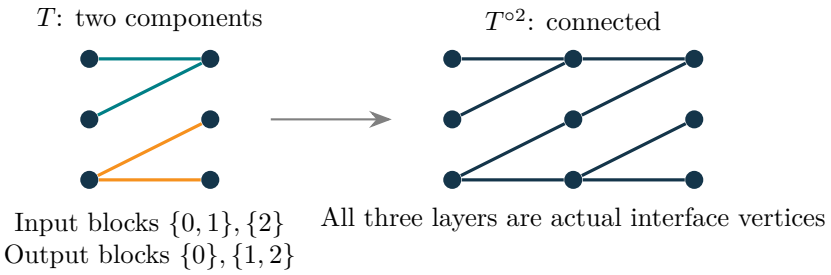


Figure 1: A through tile with initially different component partitions. After two copies the graph is connected and hence aligned. This diagram illustrates exact reblocking only; it is not a cubic terminal pattern.

$d = 2, \ldots, 8$. These finite checks test the implementation and examples; the equivalence recurrence proves the bound at arbitrary width.

The exact connected-cutoff audit compares 2,048 full signatures on 64 random connected tiles at thresholds zero through seven, covering 81,920 entries. It also checks the small-threshold sharp paths and fourteen comb examples at $k = 3, \ldots, 16$. An unrestricted independent max-flow calculation gives minimum cuts k at exponent k and $k+1$ at exponent $k+1$ for the comb boundary prescription. All checks pass. The general-threshold production constructor now uses the proved exact cutoff; the certified topological pattern query remains threshold three.

9.5 Large binary graph examples

The aligned star–triangle tile has seven vertices and six edges, interfaces $L = (0, 1, 2)$ and $R = (3, 4, 5)$, and edge list

$$(0, 6), (4, 6), (5, 6), (1, 2), (2, 3), (3, 1).$$

Its star component transports one input port to two output ports; its triangle component transports the two remaining input ports to one output. The induced component permutation is a transposition. Its closed N-copy graph has $4N$ vertices and $6N$ edges. At $N = 1$ it is K_4, supplying a positive finite graph certificate. At odd large N, the routed kernel has 36 vertices and 54 edges. For $N = 2^{4096} + 1$, the retained proof lifts its negative query answer without enumerating any missing copies.

The misaligned through example also uses $N = 2^{4096} + 1$, but needs an exact reblocking period of two first. It produces a 54-vertex, 81-edge kernel. Its source and formatted certificate occupy 1,806 and 11,384 bytes, respectively. A three-edge cut includes an original edge at copy $N - 1$, which tests the nonzero-tail address formula rather than only an early retained copy. Five source-production-plus-replay samples have median 1.519 milliseconds. This is a local binary-input demonstration; the prototype has not constructed a corresponding native knot hierarchy.

For an illustration of the avoided global period, use permutation cycles of lengths

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71.$$

The width is 639 and their global order is 557940830126698960967415390. At exponent $2^{2048} + 12345$, componentwise construction writes 132,226 vertices and 131,587 edges in 0.249 seconds in the retained run. It never writes that global order as a word. This example uses matching components and has degree two after closure; it tests only the general graph-kernel construction and makes no cubic terminal verdict.

9.6 Paired graph-query performance

Both timed arms construct their graph, answer the same cubic marked-cut predicate with the same finite certificate engine, and replay a certificate. The explicit arm expands the entire source. The routed arm constructs the bounded kernel and additionally replays its source reduction. Parsing, serialization, sphere and ball validation, and knot recognition are outside both timers. Each case has a warmup and seven paired rounds with shuffled arm order; garbage collection occurs outside each timer. The final routing run followed completion of the other in-session benchmark jobs.

Copies	Original edges	Kernel edges	Explicit (ms)	Routed (ms)	Paired ratio
1	6	6	0.108	0.208	0.53
3	18	18	0.059	0.241	0.25
9	54	54	0.103	0.380	0.27
33	198	54	0.268	0.395	0.66
129	774	54	1.023	0.363	2.86
513	3,078	54	4.281	0.377	11.04
2,049	12,294	54	17.616	0.421	42.97
8,193	49,158	54	80.700	0.459	198.94

Times are separate medians; “paired ratio” is the median of the seven within-round explicit/routed ratios, so it need not equal the ratio of the two displayed medians. The identical-explicit A/A control at 129 copies has paired median ratio 1.009. The one-, three-, nine-, and thirty-three-copy cases expose genuine reduction overhead. At 8,193 copies, the explicit median is 80.700 milliseconds and the routed median is 0.459 milliseconds. The family has a quickly found negative cut at the larger sizes; this is a complete local graph query, not a worst-case positive-flow benchmark or a comparison with the fastest possible cubic-connectivity implementation.

For exponent lengths $65, 257, 1025, 4097, 16385$ bits, the same aligned 54-edge kernel has retained query-plus-replay medians $0.293, 0.277, 0.305, 0.414, 0.610$ milliseconds. The corresponding compact JSON certificate sizes are $1768, 1960, 2728, 5800, 18088$ bytes under this benchmark’s serialization. Graph size is constant in that experiment; arithmetic, source binding, and certificate size still depend on exponent bit length. The source and certificate formatting for the separate through example is different and should not be compared byte-for-byte as an algorithmic compression ratio.

9.7 Interpretation and limitations of the measurements

The strongest measured search gain is in exhaustive multi-edge candidate generation. The clearest geometric gain is additional completed native classifications. The graph gain removes numerical multiplicity from a query whose faithful source grammar is already supplied. These are different improvements and cannot be multiplied to obtain a claimed recognition speedup.

All timings come from one shared execution environment and small deliberately specified families. Raw rounds and ranges are retained; no hardware-isolation claim is made. The proofs establish the stated asymptotic reductions under their hypotheses. The experiments establish agreement, rejection behavior, source lifting, and the local costs actually measured. A broader native workload is needed to decide the dispatch policy and identify which remaining geometric interface dominates a complete run.

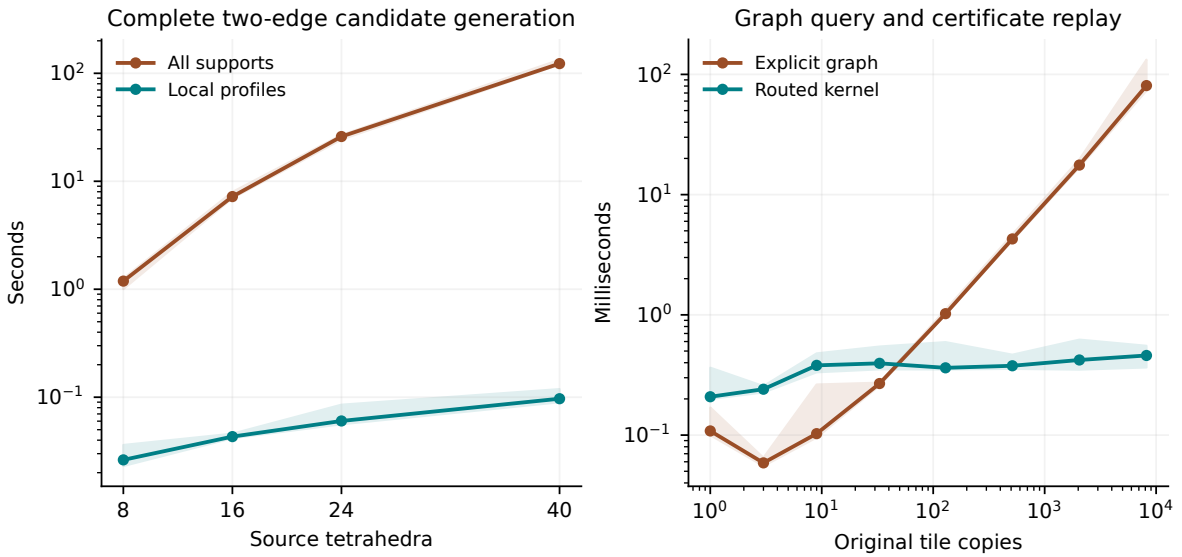


Figure 2: Two distinct local performance comparisons. Lines show retained medians and shading shows the observed sample ranges. Left: complete radius-two candidate generation on the layered obstruction family, with equal candidate sets and no topology-gate timing. Right: graph construction, query, and certificate replay on the star–triangle family. These measurements do not establish an end-to-end asymptotic knot-recognition bound.

10 What would suffice for a quasi-polynomial recognizer

10.1 An explicit parameter target for disc discovery

The logarithm of the bound in Theorem 3.3 is, up to fixed additive polynomial terms,

$$r \log t + k \log(Cd^2) + O(\log t + \log(k+1) + \log(B+1)).$$

This expression is more useful than a single claim that cancellation should be small. It identifies two different geometric questions: how dispersed the defects are, and how densely their edges meet tetrahedra.

Corollary 10.1 (A sufficient discovery budget)**.** *Suppose the encoded source sizes, coherent-coordinate bit lengths, and parameter budgets are polynomial in an original input size n. If a promised vertex meridian satisfies*

$$r \log(n+1) + k \log(d+1) = O((\log(n+1))^2),$$

then the complete cancellation-cluster search on that source is bounded by $n^{O(\log n)}$ bit operations.

Proof. Use $t \le n^{O(1)}$ in the displayed logarithmic bound. Since $d \ge 1$, $k \log(Cd^2) = O(k \log(d+1))$ for the fixed positive constant C. The remaining polynomial factors contribute $O(\log n)$ to the logarithm. □

For bounded d, the sufficient region includes $r = O(\log n)$ and $k = O((\log n)^2)$. For $d \le (\log n)^A$ it includes $k = O((\log n)^2 / \log\log(n+2))$ with the same r bound. For $d = \Theta(n)$ it requires $r + k = O(\log n)$ in this estimate. With bounded d and fixed r, even $k = O(\log n)$ yields polynomial discovery. These are conditional consequences of a proved local algorithm. No source transformation controlling these parameters on all unknot inputs is supplied.

10.2 Accounting for branching, failed attempts, and repairs

A short successful hierarchy does not bound an algorithm that searches many unsuccessful ones. The following elementary counting lemma supplies a precise way to state the missing obligation. It applies to an actual execution tree, including unsuccessful candidate choices and corrective operations.

Lemma 10.2 (Branch-budget accounting)**.** *Let a finite rooted execution tree have height at most H. For each internal vertex v, let $b(v) \geq 1$ be its actual number of children. Suppose every root-to-leaf path P satisfies*

$$\sum_{v \in P,\ v \text{ internal}} \log b(v) \leq R.$$

The tree has at most e^R leaves and at most $(H+1)e^R$ vertices. If all work charged to any one vertex is at most W, total work is at most $(H+1)e^R W$.

Proof. At every internal vertex choose a child uniformly. A leaf at path P is reached with probability $\prod_{v \in P} b(v)^{-1} \geq e^{-R}$. Leaf probabilities sum to one, proving the leaf bound. Every vertex belongs to some root-to-leaf path, each of length at most H, so counting the union of these paths with multiplicity gives at most $(H+1)$ times the number of leaves. The last assertion follows by summing the work charges. □

The height condition controls arbitrarily long unary runs. The branch sum controls actual alternatives, including abandoned choices. A restart or repair must either appear in this execution tree or have its separate cost explicitly charged. A potential that decreases only on the final successful path does not satisfy this requirement.

For example, a complete faithful implementation would meet the stronger quasi-polynomial target if it proved all of the following jointly:

1. Every reached source and all transported objects have polynomial encoded size, and every surface-search call is complete for its required topological task under the discovery budget above.
2. Every expensive terminal pattern query has a polynomial-size faithful flat descriptor whose tiles are through, or is handled by another proved operation with the same cost bound.
3. Bundle extraction, core certification, every remaining saturated attachment, and curve/disc transport take at most $n^{O(\log n)}$ work per execution node, including their certificate replay.
4. The complete execution, including failure, cleanup, and restart nodes, has $H \leq n^{O(1)}$ and $R = O((\log n)^2)$.

The local results in this report replace concrete terms in this list. Lemma 10.2 then combines the terms. The list is a research contract, not a theorem that the present recognizer already satisfies it.

10.3 The residual geometry after free-boundary condensation

The unsaturated theorem removes all new relations coming from a proper vertical attachment. Among attached bordered bases, if σ are saturated, there exists a relative decomposition with exactly σ additional two-handles from those bases. This suggests measuring complexity by a residual attaching system rather than by numerical genus or the number of product handles. It also gives a sharper diagnostic: a failure in an unsaturated region should be a source-chart

or core-certification issue, or a resource limit, for the unmarked census. New relations from these attachments occur in the saturated residue. Marked transport remains a separate issue even on an unsaturated base.

Relator count alone is insufficient. A single attaching curve can be long, can interact with many core handles, and must remain embedded with its source framing. A useful next reduction would output a small triangulated marked core and compressed boundary curves, together with a proof of their geometric meaning. The source-parallelity $30t$ bound is linear in the current source; it supplies neither a polylogarithmic residual core nor short transported attaching words automatically.

There is already a relevant algorithmic result: Chambers, Lazarus, de Mesmay, and Parsa decide contractibility of a compressed boundary curve in a triangulated three-manifold in time $2^{O(s^3)}\,\mathrm{poly}(L)$, where s is the triangulation size and L is the compressed curve description size [12]. Thus, if a future certified reduction gave a residual triangulation with $s = O(\log n)$ and $L \le n^{O(1)}$, this existing theorem would yield $n^{O((\log n)^2)}$ for that contractibility task. For $s = O((\log n)^{2/3})$ it would yield the stronger $n^{O(\log n)}$ bound. This is an algebraic substitution into a published theorem, not a new contractibility algorithm. It also concerns one specified task; finding all needed curves and recognizing the resulting manifold require additional work.

10.4 Why the pattern result changes the remaining target

The incoming disconnected-pattern question required control of a potentially large global permutation order. Componentwise routing removes that order. Reblocking removes the need for initially aligned partitions whenever every component meets both interfaces. Consequently a faithful flat through-tile descriptor is now sufficient; there is no need to force every original tile to be connected or to have logarithmic interface width.

This enlarges the target class for a geometric producer in a precise way. The remaining obstructions are components missing an interface, access to private tile vertices by later geometry, and unrestricted nested reuse. These are concrete structural properties that can be logged at the native pattern-transport boundary. Measuring those failures is more informative than inferring applicability from a large numerical parallelity multiplicity alone.

11 Further research questions and a proposed program

The results suggest a research program centered on geometric interfaces and explicit parameter budgets. The priorities below are ordered by their ability to connect the proved local reductions to a complete recognition procedure. Each asks for a mathematical statement or a reproducible artifact that can be accepted or falsified independently.

11.1 Priority I: close the native geometric interfaces

1. A faithful through-tile producer. Can the source-parallelity charts and ordered cut-boundary words produce a flat descriptor of the actual transported cubic pattern in polynomial encoded time? The new theorem allows disconnected tiles and initially different component partitions. A first target is a nontrivial complete class of normal cuts with a proof that every tile component meets both interfaces. The producer must record the spherical embedding, actual port identities, edge ownership, and why subsequent context has no access to private vertices. A useful experiment logs these facts and each precise rejection reason on native hierarchy cuts; a count of large multiplicities alone tests none of them.

2. Compressed relative bands with marked transport. Can the proper-boundary handle theorem be made constructive at the level of source charts without writing one band per unit of genus? Seek a compressed relative spine whose expansion gives the actual band system and whose actions on ordered boundary arcs are polynomial in the encoded input. One should specify which operations on that spine remain polynomial: intersection, concatenation, extraction of a co-core disc, and transport of a boundary pattern. A scalar genus counter settles the unmarked census but is insufficient for any operation that must return an embedded curve or disc.

3. Compile the saturated residue geometrically. Each saturated bordered base contributes one two-handle in a suitable relative decomposition. Can its attaching curve be encoded directly from the ordered interface words and bundle charts, with a short embedded certificate? A natural first family is planar saturated bases with a bounded number of interfaces, followed by orientable bases with large binary genus and then nonorientable bases. The output must preserve the attaching map and framing; recording only its abelianization loses the distinction exhibited by $\langle x, y \mid x^3 = y^2 \rangle$. A successful theorem should bound both the number of residual curves and their encoded interaction with the core.

4. Extract a source-addressed geometric noose. The graph kernel gives at most three original crossing edges and a compressed shore coloring. Can the actual spherical embedding turn this into a simple closed curve with a compressed isotopy certificate, then transport the corresponding disc through the hierarchy? This is an appropriate use of existing efficient algorithms for curves on surfaces [11]. The extra issue is the source provenance linking those surface calculations to the three-dimensional cut and its successive boundary identifications. The first deliverable should cover a repeated native product region, including reversed charts and corners.

11.2 Priority II: prove discovery and execution budgets

5. Bound the cancellation component count on useful sources. The fifteen-tetrahedron example proves that a one-component assumption cannot be imposed on every vertex meridian. Does some broader small bound hold after a certified source transformation, or can one construct a family in which the minimum possible component count tends to infinity? A convincing negative family must control *all* vertex meridians, rather than exhibit one complicated meridian. A positive theorem should bound the expression $r \log n + k \log(d + 1)$, which is what the discovery algorithm actually pays. This is a more flexible target than independently minimizing all three parameters.

6. Control incidence without losing the one-vertex model. Can certified moves reduce the maximum tetrahedron-incidence degree d while keeping t and the required cancellation budget under control? Subdivision can lower local incidence in some representations but may introduce vertices; the fixed-cocycle proof then stops applying. The desired statement must quantify this tradeoff and preserve the marked exterior. A useful intermediate theorem would bound the sum of local profile costs over all affected tetrahedron stars after a Pachner move, rather than restart every profile search on the whole source.

7. Several vertices with shared potentials. For multiple vertices, write a representative as $c + \delta\phi$ and account for the sign chambers of its edge values. Can local cancellation profiles be coupled through a small exact system in the shared vertex potentials? A Schur-complement

or arrangement-based approach is plausible when the number of independent potentials is small. The proof must retain integer feasibility and the correct boundary-cohomology quotient. A real-valued local solution or an arbitrary fixed vector $|c|$ would not be a complete substitute. The earlier sign-chamber machinery provides a baseline against which a new bound should be compared [3].

8. Reduce the active-system dependence on incidence. The factor $(Cd^2)^k$ counts connected supports and then local active hyperplane choices. Could these two enumerations be combined through an incidence graph of equations and edges? A potential gain is a smaller dependence on d or a bound in the number of distinct local coefficient patterns. The difficulty is that an active-system incidence graph can disconnect even when the cancellation support is connected. Any pruning rule must retain all extremal profiles, including the homogeneous exceptional case; the current complete algorithm is an exact oracle for small tests.

9. Independently certify restricted exhaustion. The positive rank gate is compact. A general negative search transcript is not yet supplied. Can one certify that every connected support and every relevant active system up to the chosen budgets has been accounted for, sharing repeated coefficient patterns and exact row reductions? The one-edge inventory for the retained counterexample is a tractable starting case. Such a transcript would establish exhaustion of a *restricted* disc class. It would become a nontriviality certificate only after a separate theorem makes that class sufficient for all required meridians.

10. An amortized hierarchy potential for all work. Which geometric potential bounds both useful cuts and corrective operations, while a branch budget controls all alternatives actually explored? A candidate must be tested against source rebuilding, pattern normalization, saturated repairs, and failed discovery. The execution-tree statement in Lemma 10.2 provides an explicit acceptance criterion. Instrumenting these charges is useful before proving a global bound: a local strict descent that triggers expensive unrestricted restarts may have little practical value despite its mathematical validity.

11.3 Priority III: enlarge and improve the graph kernels

11. Tiles with births, deaths, and private components. The through hypothesis excludes a component that meets only one interface or neither. Is there a polynomial marked-cut kernel for every binary tile power at a fixed threshold? One possible route uses the larger partition-diagram semigroup and records components that become internal during composition. Purely private repeated connected components need only a bounded number of representatives to supply the two marks, but components that enter and later leave a through region require a source-addressed transient analysis. A proof must track those components; the rank recurrence of Lemma 6.1 deliberately does not cover them.

12. Can the quadratic reblocking bound be reduced? The exponent bound $m \le d$ and the universal connected cutoff are sharp. They do not show that a d^2Q kernel is necessary. The sharper estimate $(a_k+1)\sum_{\mathcal{O}} h_{\mathcal{O}} Q_{\mathcal{O}} + (m-1)Q$ already distinguishes easy instances. Can shared cycle structure give an $O(Q)$ or $O(Q\log Q)$ explicit marked representative, or is a superlinear lower bound forced by exact boundary marks and source lifting? Known terminal mimicking networks suggest comparison tools [16, 17], but their construction from an expanded graph does not by itself solve the compressed input problem.

13. Stable kernels under nested descriptions. Flat powers have explicit tiles. A general straight-line grammar may duplicate already compressed subgraphs, so repeated local kernelization can increase intermediate size. Identify a grammar class closed under the required geometric operations for which kernel size is bounded by a fixed polynomial in the whole grammar. A useful first result would charge each original edge definition only a logarithmic number of times. The proof must preserve exposed vertex identity and exact mark unions at each composition, not only the final yes/no answer.

14. Smaller positive graph proofs. The dense replay path intentionally makes a conservative quartic pass over the finite kernel. A sparse replay can remove an avoidable factor, and positive anchor-pair proofs may share cut or flow structure. The mathematical question is whether a short family of certified witnesses implies coverage of all anchor pairs without enumerating them individually. The practical question is where that proof work dominates source construction on native patterns. Changes should be measured on complete producer-plus-replay queries, including small instances where reduction overhead loses.

11.4 A concrete sequence of research deliverables

1. **Native interface census.** Instrument a fixed corpus of actual hierarchy cuts. Record (t, k, r, d) where measurable, saturated versus proper base attachments, faithful tile sizes, through failures, and all source and replay costs. Retain inconclusive outcomes and exact source identities.
2. **One complete geometric adapter.** Choose the largest clearly specified family in that census for which a faithful pattern grammar and relative band transport can both be proved. Deliver a source-bound producer and independent replay, then measure completed recognition stages.
3. **One global parameter theorem or obstruction family.** Establish a sufficient discovery budget on a certified family, or construct a family forcing its violation. Either result would materially refine the strategy; more favorable timings without such structural information would not.
4. **A complete execution account.** Integrate the proved adapters with the maintained recognizer, charge every failed attempt and repair, and state the final complexity theorem with all input-size conversions. Until that account is complete, retain the local results as proved reductions and their global use as an explicit research objective.

A Reproduction, dependency provenance, and file inventory

The archive is self-contained for the three standard-library prototype suites and the retained certificate replays. Run from its root:

```
python3 verify_all.py
PYTHONPATH=code python3 -m routed_patterns \
  examples/through_4097_bits.json --through \
  --output reproduced_through_certificate.json
PYTHONPATH=code python3 -m routed_patterns \
  examples/through_4097_bits.json \
  --verify examples/through_4097_bits_certificate.json
python3 experiments/reblocking_partition_audit.py \
```

```
  --max-width 5 --output results/repeated_partition_audit.json
python3 experiments/connected_cutoff_audit.py
cd article
latexmk -pdf -interaction=nonstopmode hierarchy_kernels.tex
```

The verification runner isolates the three import environments, so the different incoming dependency snapshots cannot accidentally replace one another. It records the observed test counts and all saved-certificate replays. The cancellation package defaults to its bundled runtime; environment overrides can instead point to an explicitly prepared repository checkout. The geometry package contains only its pinned transitive upstream modules and the retained corpus, plus minimal reproduction package namespaces.

The recorded interpreter is CPython 3.12.14 on Linux x86-64. The production code uses exact integers and rational arithmetic from the standard library. Regina 7.4 is optional for the expanded geometric and full vertex-enumeration oracles [13]; it is not needed for replaying the retained positive certificates. Matplotlib is used only to regenerate the report's static performance figure. The article builds with pdfLaTeX and ordinary TeX Live packages. Long full-enumeration benchmarks are separate from the quick verification command and should be run without simultaneous compute-heavy jobs.

Path	Role
`article/hierarchy_kernels.tex`, `article/sections/`	Main article, complete theorem proofs, experiment interpretation, and research program
`article/hierarchy_kernels.pdf`	Compiled and visually checked report
`article/connected_cutoff_supplement.tex`	Exact cutoff theorem and sharp constructions at every threshold
`code/routed_patterns/`	New aligned routing, exact cutoff, through-component reblocking, and independent checker
`code/pattern_kernel/`	Unchanged incoming tile types, finite graph certificate engine, replay, and small full-signature oracle
`cancellation/fastunknot/`	New connected-profile discovery module and unchanged incoming positive vertex-disc gate
`cancellation/runtime/`	Minimal pinned search dependencies and the coherent-obstruction source fixture
`cancellation/artifacts/`	Exact disconnected disc, all external vertex vectors, one-edge inventory, tests, and paired timing records
`geometry/code/unsaturated_bundles/`	Finite proper-boundary classifier, independent census replay, and native source adapter
`geometry/vendor/`, `geometry/evidence/`	Pinned incoming runtime, 106-case corpus, five newly completed source certificates, and measurements
`examples/`	Four graph sources and independently replayable proofs, including the large misaligned through-component example
`experiments/`, `results/`	Full-signature and partition audits, source-coloring comparisons, timing drivers, raw results, and review records
`verify_all.py`	Quick isolated test and saved-certificate replay runner

Path	Role
`integration/INTEGRATION.md`	Additive repository placement and exact native adapter contracts
`provenance/`	Source checkpoint and copied-dependency identities

The archive deliberately includes enough pinned source to reproduce the results without a live branch moving underneath them. Those copied modules are not counted as new implementations. No maintained checkout changes, remote repository writes, full native regression-suite run, or end-to-end knot-recognition performance result is implied by this package.

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